Therapeutic Interventions for Gut Dysfunction:

The 5R Approach

JENNIFER JACKSON, MD

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Jennifer Jackson, MD, disclosed no relevant financial relationships with any ineligible companies.

Robert Luby, MD, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. List the components of the 5R therapeutic approach.
2. Align each of the 5R interventions with their most closely associated DIGIN gut dysfunction.
3. Explain the rationale for the primacy of Remove and Rebalance interventions when sequencing the 5R therapeutic approach.
4. Explain why it is often necessary to apply all five of the components of the 5R approach and why many of the therapeutic agents should be considered as temporary “bridge” interventions.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Initiate interventions that Remove and Rebalance dietary factors and lifestyle behaviors that mediate physiological dysfunction as initial therapeutic interventions for all DIGIN gut dysfunctions.

2. Initiate interventions to Replace deficient digestive factors, Repopulate the gut microbiome, and Repair increased intestinal permeability when clinically indicated.

Foundational Therapeutic Approach

Gastrointestinal – Assimilation

LIFESTYLE FIRST

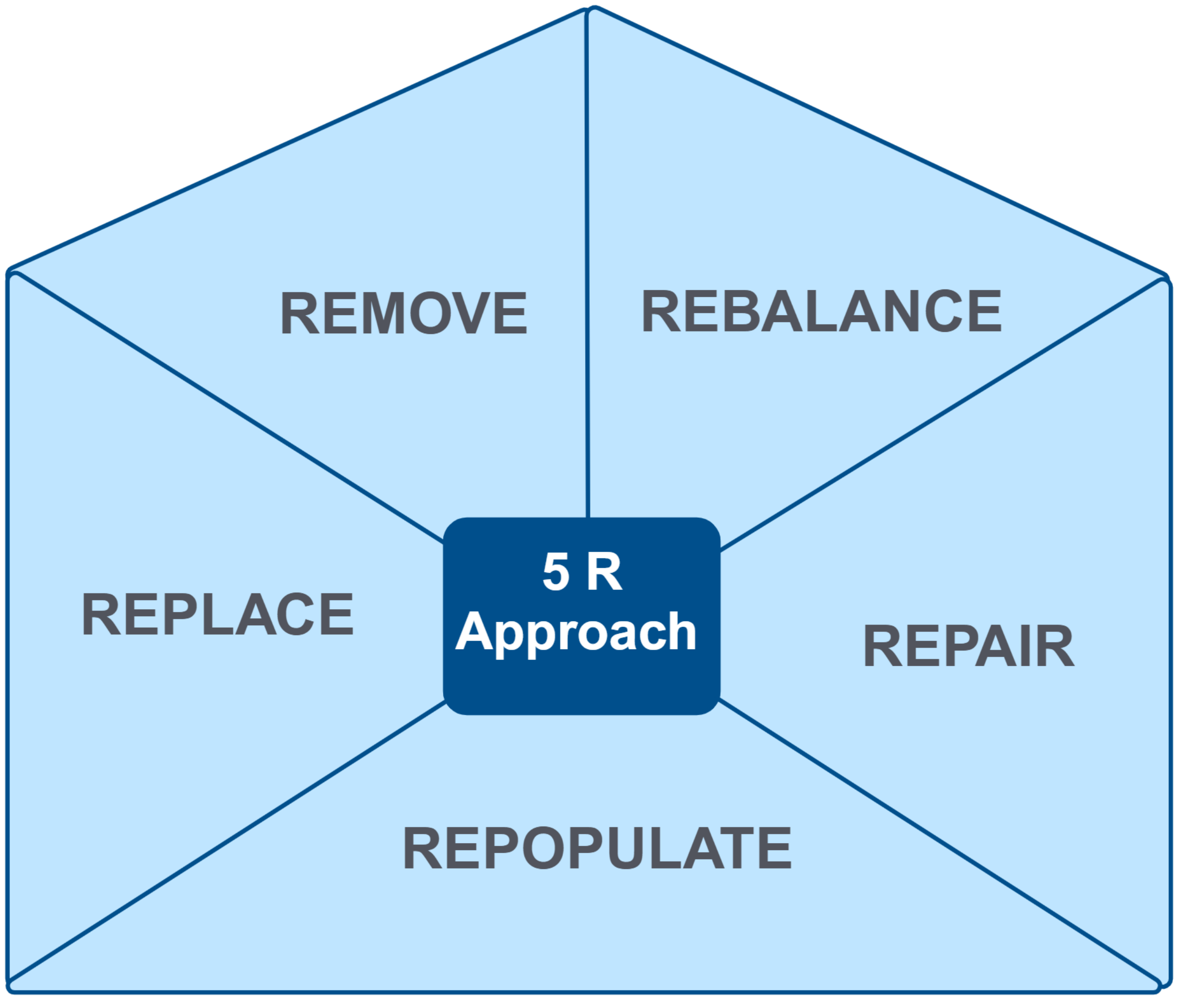
FOOD FIRST

GUT FIRST

Clinical Focus

- **<u>Remove</u>** interventions for DIGIN dysfunctions.

Therapeutic Interventions for Gut Dysfunction: The 5R Approach



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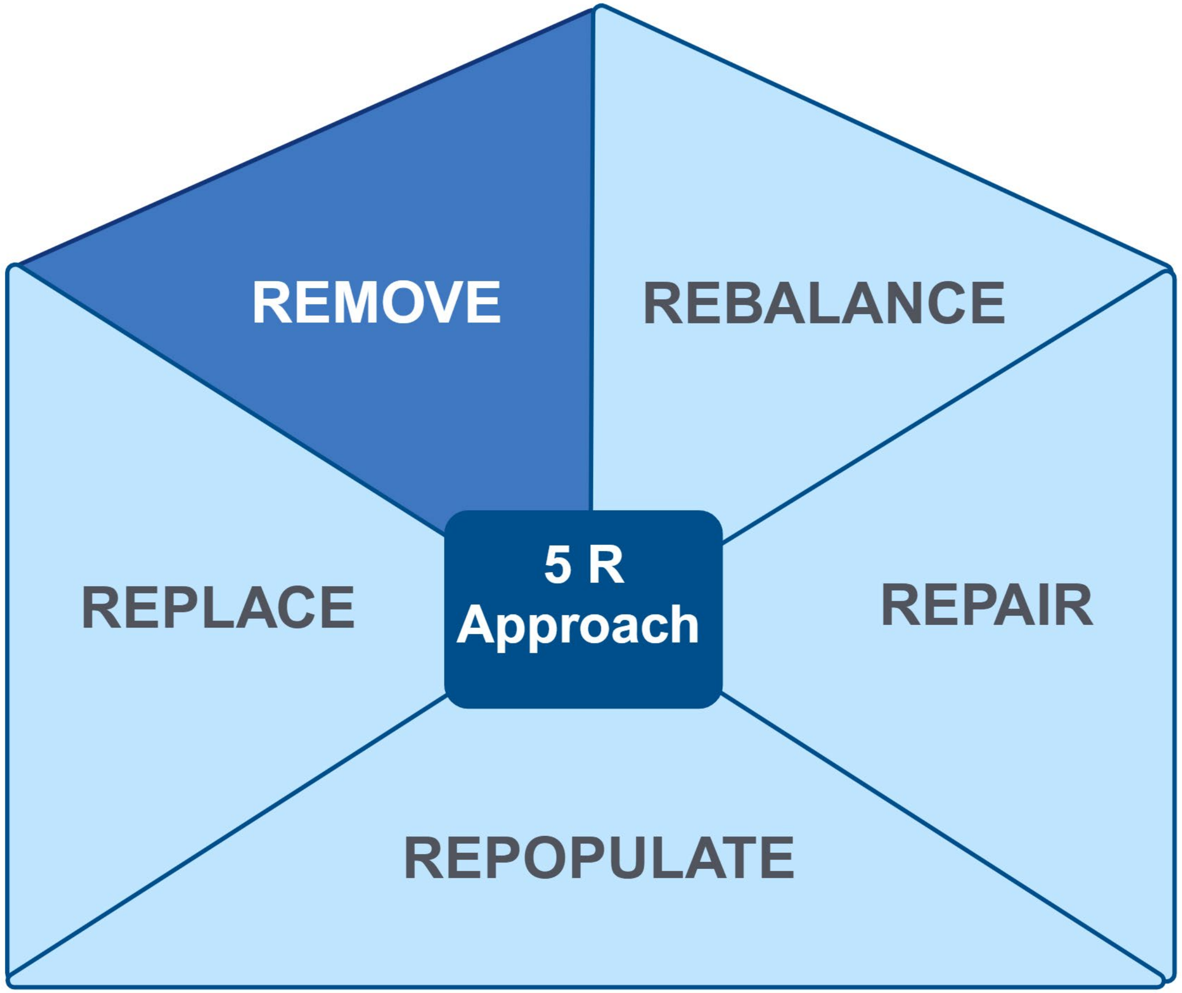
Long Description: Therapeutic Interventions for Gut Dysfunction: the 5 R Approach

- The 5 R Approach includes remove, replace, repopulate, repair, rebalance.

Therapeutic Interventions for Gut Dysfunction: The 5R Approach: Remove



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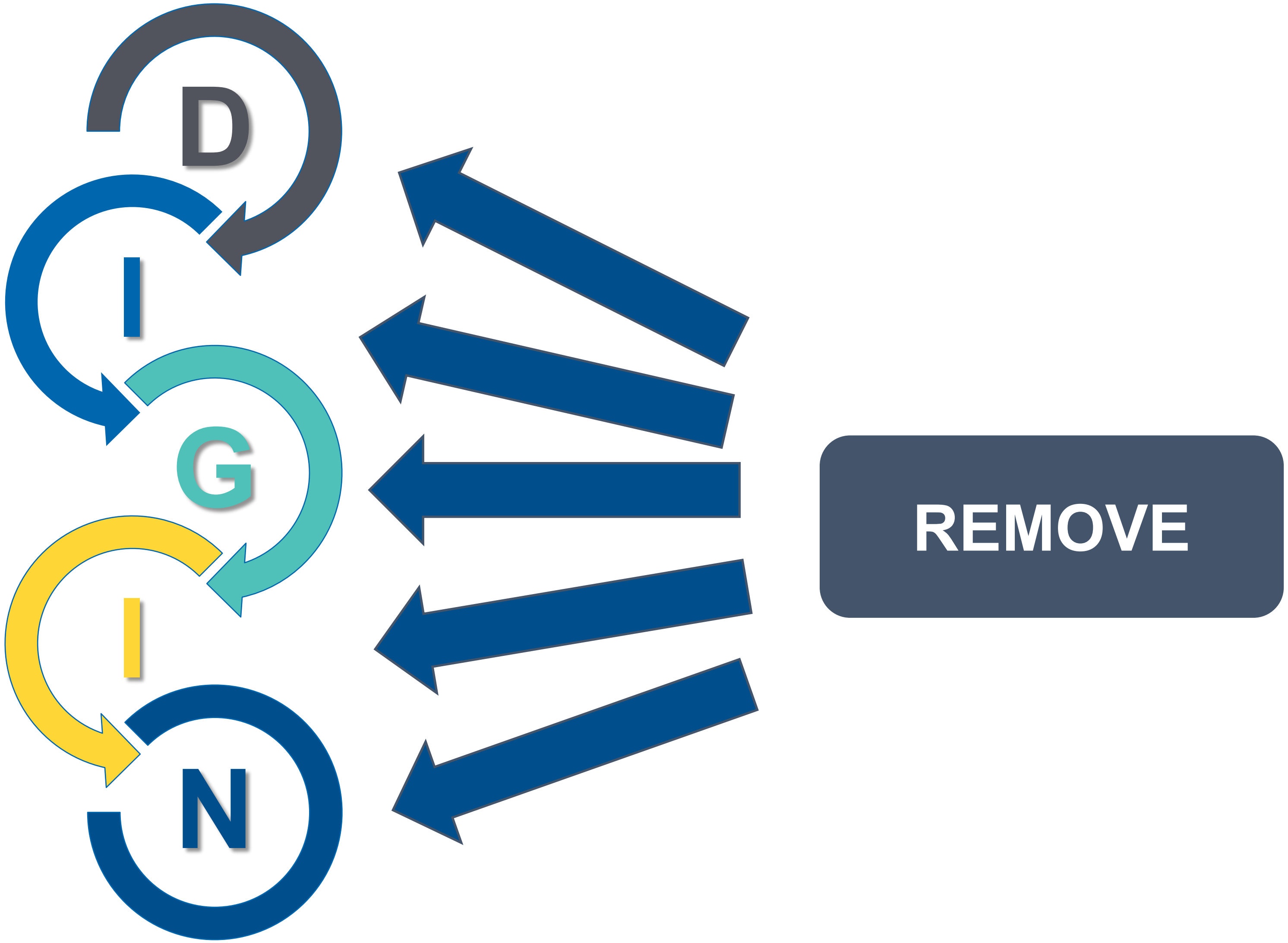
Long Description: Therapeutic Interventions for Gut Dysfunction: the 5 R Approach

- The 5 R Approach includes remove, replace, repopulate, repair, rebalance. Here the emphasis is on remove.

Remove Interventions
Effective for All Five DIGIN Dysfunctions



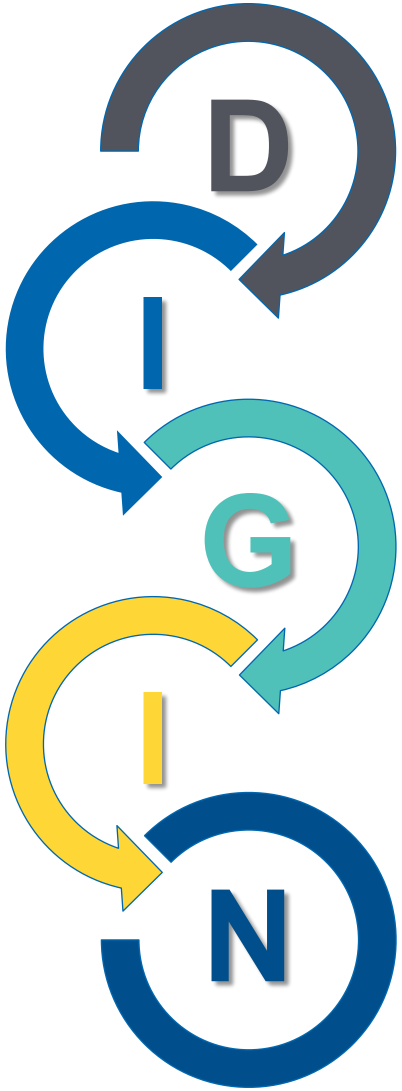
The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Long Description: Remove Interventions

- The Remove phase of the 5 R Approach addresses all five of the D I G I N dysfunctions.

Foundational **Remove** Interventions

Foods – Drugs – Bugs – Behaviors



- D – PPIs, antacids, metformin, anti-nutrient foods
- I – Alcohol, refined/processed carbs, NSAIDs, stress
- G – Problem foods, antibiotics, pathogenic/dysbiotic microbes
- I – Ultra-processed foods; (artificially) sweetened beverages; stress; single foods that invoke allergy, sensitivity, intolerance (gluten, dairy, etc.)
- N – Foods that impair motility, adverse lifestyle/meal behaviors

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Clinical Pearl

Rapid Identification of **Remove** Interventions

HOW WOULD YOU COMPLETE THESE STATEMENTS?

My health would be better if I ate less _______________.

My health would be better if I drank less _______________.

HOW WOULD YOU COMPLETE THESE STATEMENTS?

My health would be better if I spent less time _______________.

My health would be better if I spent more time _______________.

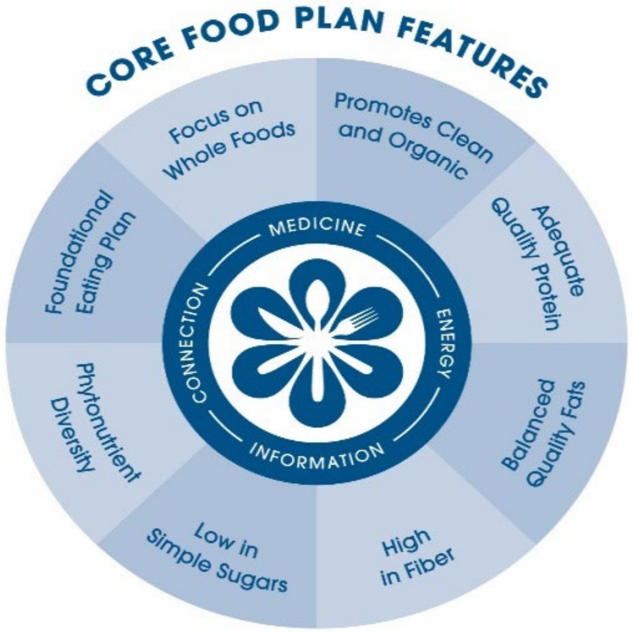
The Clinical Path to the Foundational Food Plans

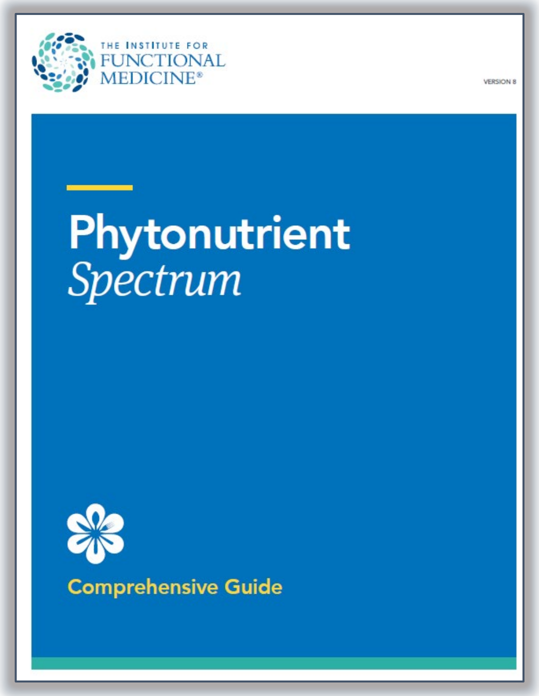
Medical History

Chief Complaints
Conditions
Timeline and ATMs
Medication Review

ABCDs of Nutrition Evaluation

Anthropometrics
Biomarkers and Labs
Clinical Indications From NPE
Diet and Lifestyle Review
Matrix Review





Accessible version available in your IFM Toolkit.

Long Description: The Clinical Path to the Foundational Food Plans

- The Core Food Plan and the Phytonutrient Spectrum are Toolkit Items. The Core Food Plan emphasizes how food is medicine, energy, information, and connection. Its features include: focus on whole foods, promote clean and organic foods, adequate quality protein, balanced quality fats, high in fiber, low in simple sugars, phytonutrient diversity, foundational eating plan

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Nutrition Interventions

Functional Nutrition Fundamentals

Food Is...

- Energy
- Information
- Connection
- Medicine

Foundational Interventions

- Phytonutrient Spectrum Foods
- Phytonutrient Spectrum Checklist
- Core Food Plan
- Core Food Plan–Vegan
- Core Food Plan–Vegetarian

Therapeutic Interventions

- Cardiometabolic Food Plan
- Elimination Diet
- Detox Food Plan
- Mito Food Plan

Personalized Nutrition Plan

Long Description: Nutrition Interventions

- Functional Nutrition Fundaments:
- Food is: energy, information, connection, medicine
- Foundational Interventions include the following tool kit items: Phytonutrient Spectrum Foods, Phytonutrient Spectrum Checklist, Core Food Plan, Core Food Plan Vegan, and Core Food Plan Vegetarian
- Therapeutic Interventions include the following tool kit items: Cardiometabolic Food Plan, Elimination Diet, Detox Food Plan, Mito Food Plan
- Practitioners can utilize these tools to create a personalized nutrition plan for the patient

The Core Food Plan

Recommended "Core" First Step for Most Patients

A More Nutrient-Dense Diet

- Inflammatory elements
- Inconsistent behaviors
- Imbalanced fats
- High-glycemic impact
- Ultra-processed

Nutrient-Depleted Diet

Core Food Plan

- Nutrient-dense
- Healthy whole foods
- Phytonutrient rainbow
- Balanced dietary choices

Connection to the Assimilation Node

Features include the removal of inflammatory foods while encouraging the inclusion of foods that balance the microbiome and support digestion.

Long Description: Recommended “Core” First Step

- Many patients have a nutrient-depleted diet containing inflammatory elements, inconsistent behaviors, imbalanced fats, high-glycemic impact, and ultra-processed foods. Comparatively, the Core Food Plan is nutrient-dense, contains healthy whole foods, uses the phytonutrient rainbow, and has balanced dietary choices. The Core Food Plan also connects to the assimilation node because it removes inflammatory foods and encourages foods that balance the microbiome and support digestion.

The Problem, the Solution, and the Goal

- Low-grade inflammation plays a key role in the pathophysiology of gut dysfunction.
- Several of the Core Food Plan features specifically target reducing inflammation and addressing food as a mediator of gut dysfunction.
- With food eliminations, the patient is removing food antigens that may be the cause of the inflammation or may become a problem as the result of inflammation. The Core Food Plan also eliminates addictive and potentially toxic or harmful foods.

As the patient restores gut function, it may be possible to successfully reintroduce the "problem foods."

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Nutrition Interventions

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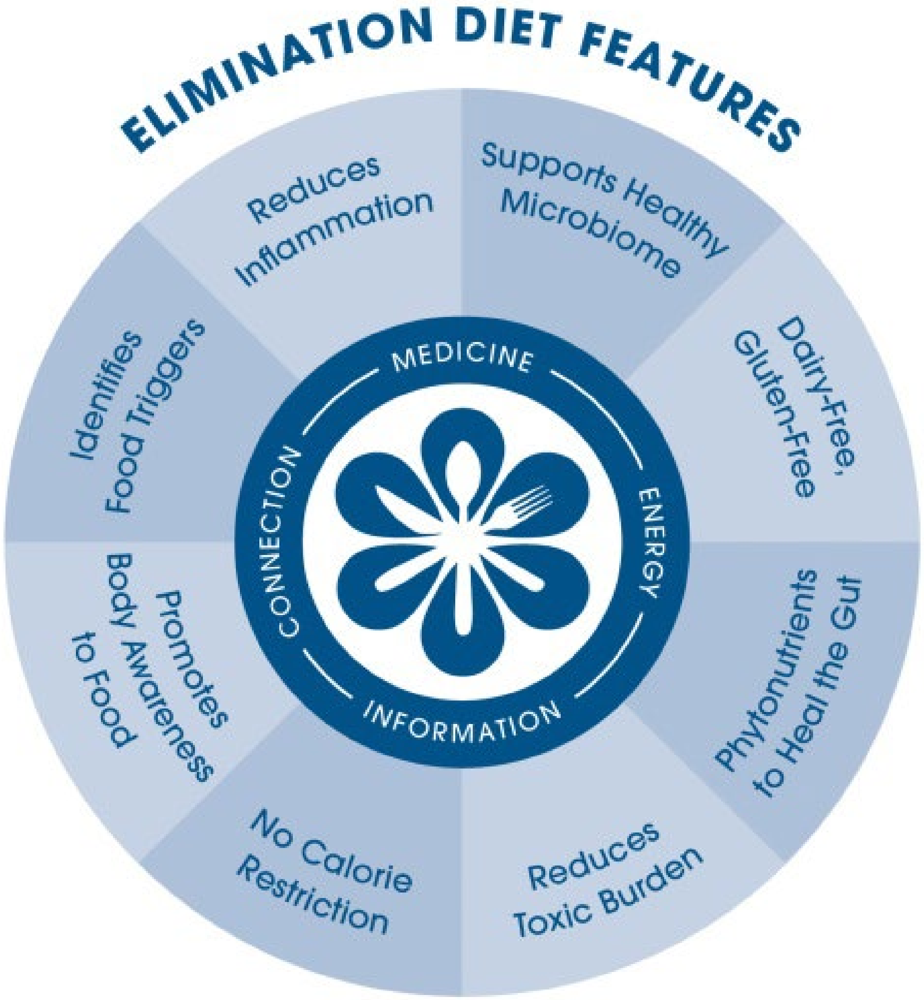
What Is IFM's Elimination Diet?

A moderately restrictive, short-term approach followed for a specific amount of time, often used for:

- Ridding the body of the most common foods that cause inflammation.
- Addressing clinical suspicion of adverse food reactions and food as a mediator of any of the DIGIN dysfunctions.
- Eliminating foods with potentially addictive, toxic, or harmful components.

IFM's Elimination Diet Features

1. Identifies Food Triggers
2. Reduces *[more]* Inflammation
3. Supports Healthy Microbiome
4. Dairy-Free, Gluten-Free
5. Phytonutrients to Heal the Gut (Phyto Spectrum)
6. Reduces Toxic Burden
7. No Calorie Restriction
8. Promotes Body Awareness to Food



Long Description: I F M’s Elimination Diet Features

- The Elimination Diet emphasizes food as medicine, energy, information, and connection. Its features include: Identifies Food Triggers, Reduces Inflammation, Supports Healthy Microbiome, Dairy-Free, Gluten-Free, Phytonutrients to Heal the Gut, Reduces Toxic Burden, No Calorie Restriction, Promotes Body Awareness to Food

Diets That Ameliorate Gut (DIGIN) Dysfunctions

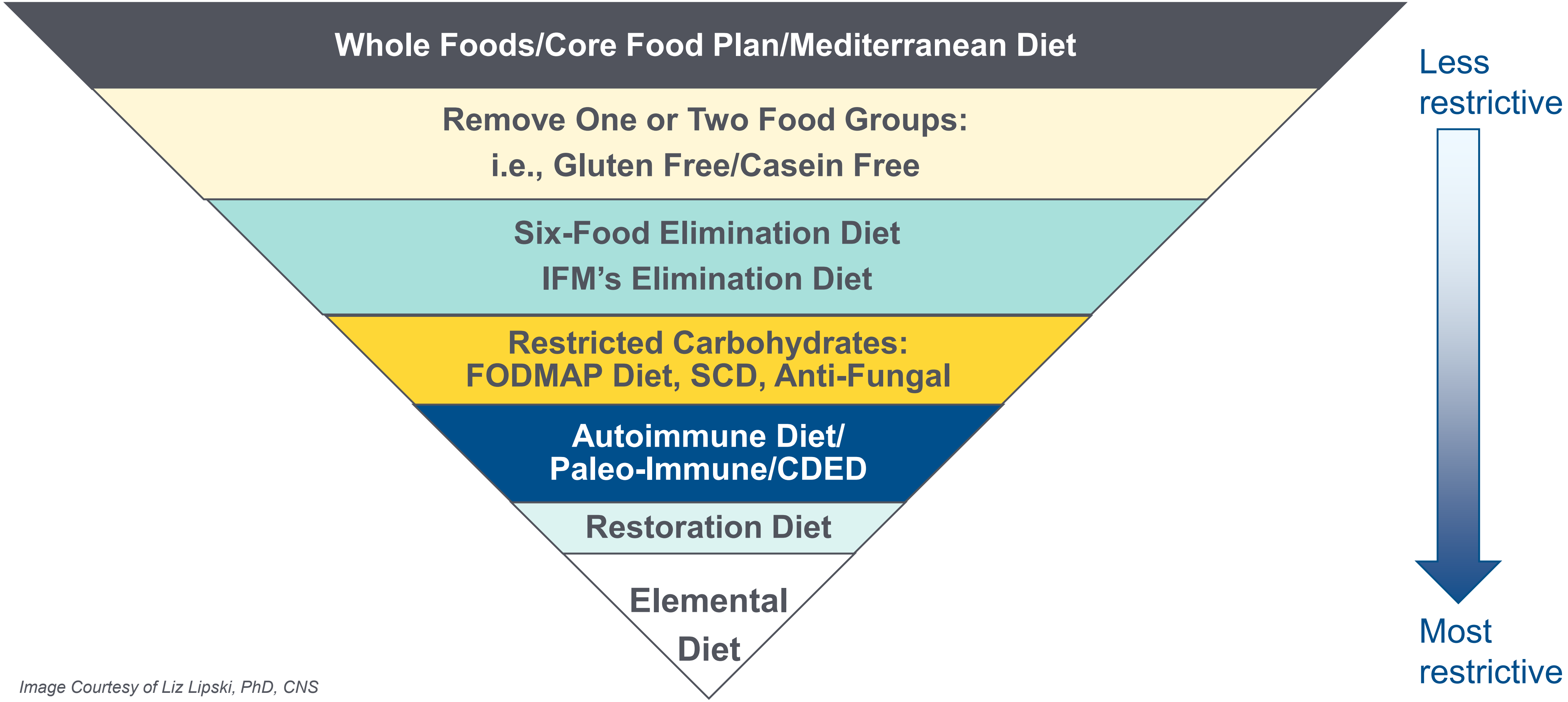


Image Courtesy of Liz Lipski, PhD, CNS

Long Description: Diets That Ameliorate Gut (D I G I N) Dysfunctions

- This image depicts an inverted triangle. At the widest part of the triangle are the least restrictive diets and food plans including whole foods, the IFM core food plan, and the Mediterranean diet. As you travel down the triangle to the point, the diets become more restrictive. The next level includes removing one or two food groups such as gluten free or casein free. Next is a six-food elimination diet and IFM’s Elimination Diet. Becoming more restricted are the restricted carbohydrate diet, FODMAP Diet, S C D, and anti-fungal. Next includes the Autoimmune Diet, Paleo-Immune, and C D E D. Next is the Restoration Diet, and the most restrictive diet at the tip of the triangle is the Elemental Diet.

Core Food Plan & Elimination Diet Features

CONNECTION BETWEEN FOUNDATIONAL AND THERAPEUTIC FOOD PLANS

Core Food Plan & Elimination Diet	How it improves DIGIN dysfunctions
Focus on Whole Foods & Phytonutrient Diversity	Phytonutrients to Heal the Gut, Supports Healthy Microbiome, Reduces Inflammation
Encourages Organic	Reduces Toxic Burden
Adequate Quality Protein	Reduces Inflammation
Balanced Quality Fats	Reduces Inflammation
High Fiber	Reduces Inflammation, Supports Healthy Microbiome, Reduces Toxic Burden
Low in Simple Sugars	Reduces Inflammation
Dairy-Free, Gluten-Free (optional)	Reduces Inflammation, Supports Healthy Microbiome, Reduces Toxic Burden

What Is Already Removed in the Core Food Plan?

Feature	What is naturally removed?
Focus on Whole Foods & Phytonutrient Diversity	*Removes high-calorie, low-nutrient, ultra-processed foods; limits alcohol and caffeine*
Encourages Organic	*Limits/removes GMOs, reduces toxin exposure*
Adequate Quality Protein	*Removes low-quality, high-fat proteins*
Balanced Quality Fats	*Removes imbalanced, inflammatory fats, including eliminating trans, hydrogenated, and partially hydrogenated fats (typically found in ultra-processed foods); decreasing saturated fats from animal sources; increasing omega-3 fats from fish and plant sources*
High Fiber	*Removes refined grains*
Low in Simple Sugars	*Removes high-glycemic impact, including sugary beverages, processed foods, refined grains, added & artificial sweeteners*

Considerations & Contraindications

Indications for Referral

Indications to Consider Referral to a Qualified Nutrition Specialist

- Elimination Diet
- Cardiometabolic Food Plan
- Detox Food Plan
- Mito Food Plan
- Low FODMAPs Diet
- Anti-Candida Food Plan
- All GI-Specific Diets
- All "Restricted" Diets: Paleo, Carnivore, Pesca/Vegetarian, Vegan

- Atia O, Shosberger A, Focht G, et al. Development and validation of the IBD-REFER criteria: early referral for suspected inflammatory bowel diseases in adults and children. *Crohns Colitis 360*. 2020;2(2):otaa027. doi:10.1093/crocol/otaa027
- Image: Created with BioRender.com by The Institute for Functional Medicine

Restrictive/Elimination Diet Red Flags

Indications for Referral

- **History of anaphylaxis**: Must work with a qualified practitioner who can diagnose and treat food allergies. Elimination diet may contribute to loss of oral tolerance.
- **Atopic conditions**: Proceed with caution, especially in those with atopic dermatitis and history of food anaphylaxis. Higher risk of developing an allergy to previously tolerated foods and anaphylaxis to food after an elimination diet.
- **Food protein-induced enterocolitis syndrome (FPIES)**: Severe reaction during reintroduction is possible. Ensure working with a qualified practitioner who can diagnose and treat FPIES.

- Image: Created with BioRender.com by The Institute for Functional Medicine
- See: “References: Restrictive/Elimination Diet Red Flags”

Considerations & Contraindications
Indications for Referral

- **Nutrition insecurity or nutrient deficiencies**: Assess nutritional status and work with qualified practitioners.
- **Populations with specific nutritional needs**: Including children, pregnancy, lactation, or other populations with varying calorie needs and individuals on "restricted" diets: paleo, carnivore, pesca/vegetarian, vegan. Work with a qualified nutrition professional.
- **History of or current eating disorder; dysfunctional relationship with food (ex: orthorexia)**: If assessed and indicated, care team should consist of qualified mental health practitioner(s), PCP, and a qualified nutrition professional.

- Image: Created with BioRender.com by The Institute for Functional Medicine
- See: “References: Restrictive/Elimination Diet Red Flags”

References

RESTRICTIVE/ELIMINATION DIET RED FLAGS

- Atia O, Shosberger A, Focht G, et al. Development and validation of the IBD-REFER criteria: early referral for suspected inflammatory bowel diseases in adults and children. *Crohns Colitis 360*. 2020;2(2):otaa027. doi:10.1093/crocol/otaa027
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RESTRICTIVE/ELIMINATION DIET RED FLAGS

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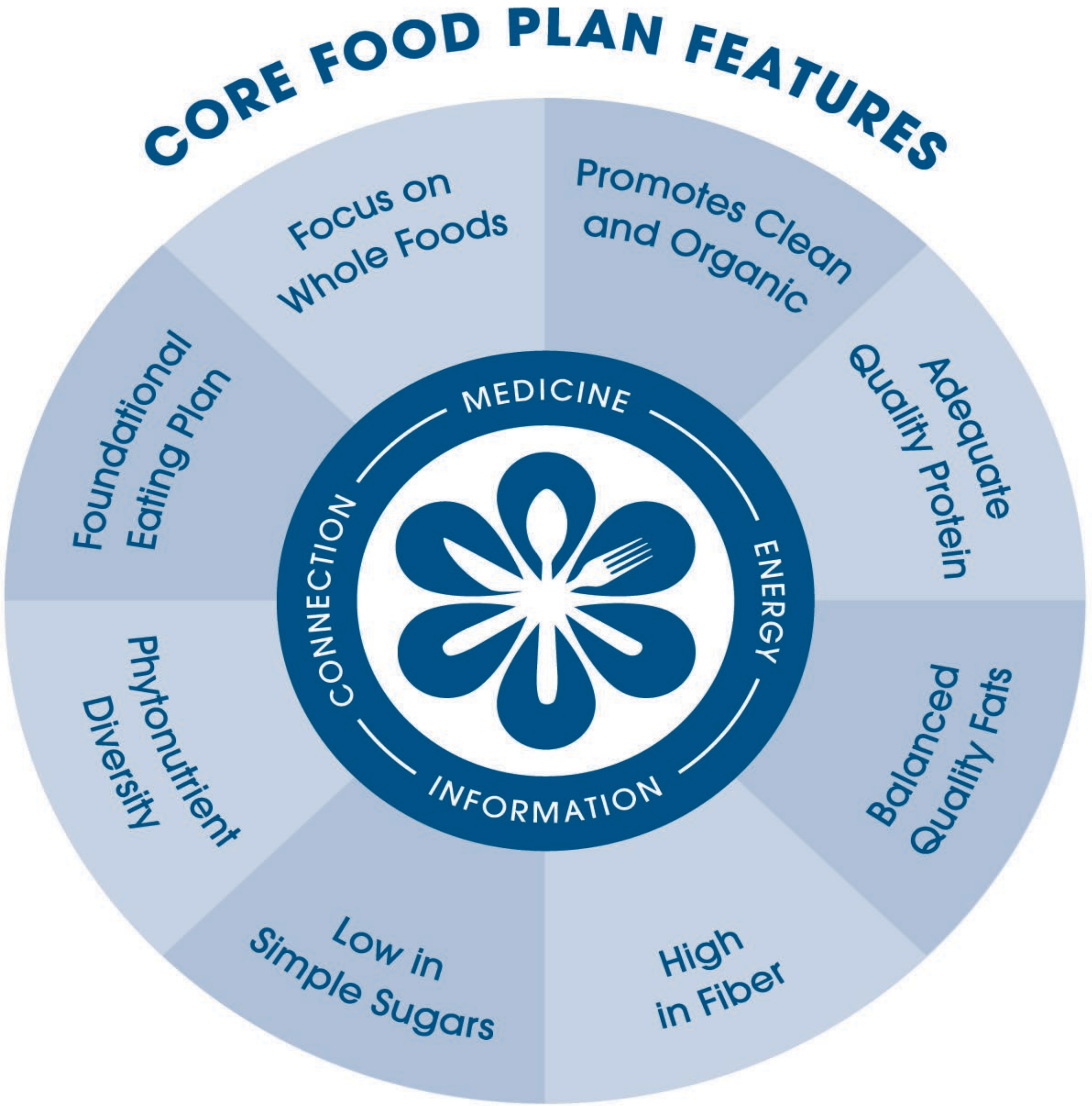
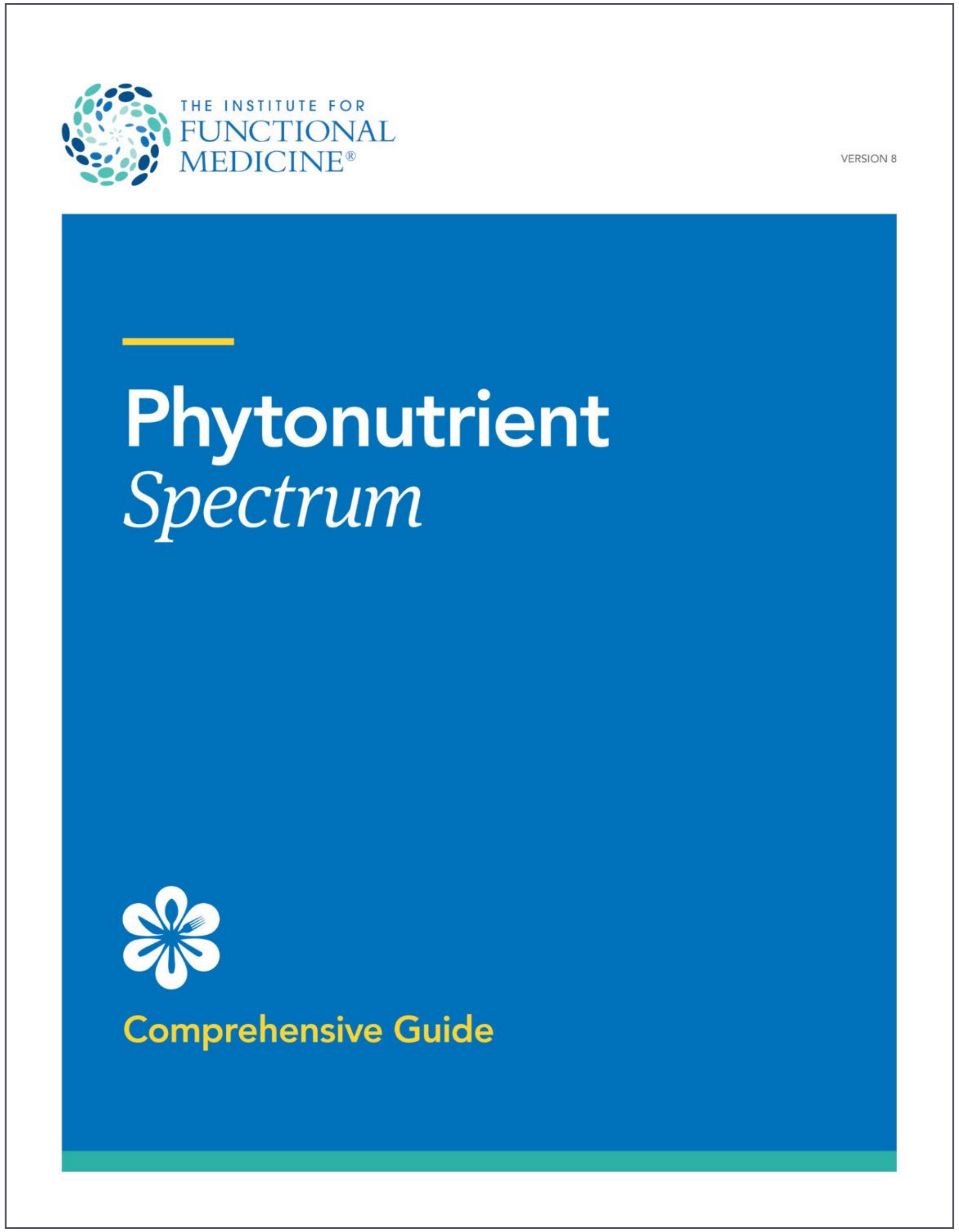




Accessible version available in your IFM Toolkit.

Long Description: Core Food Plan Features

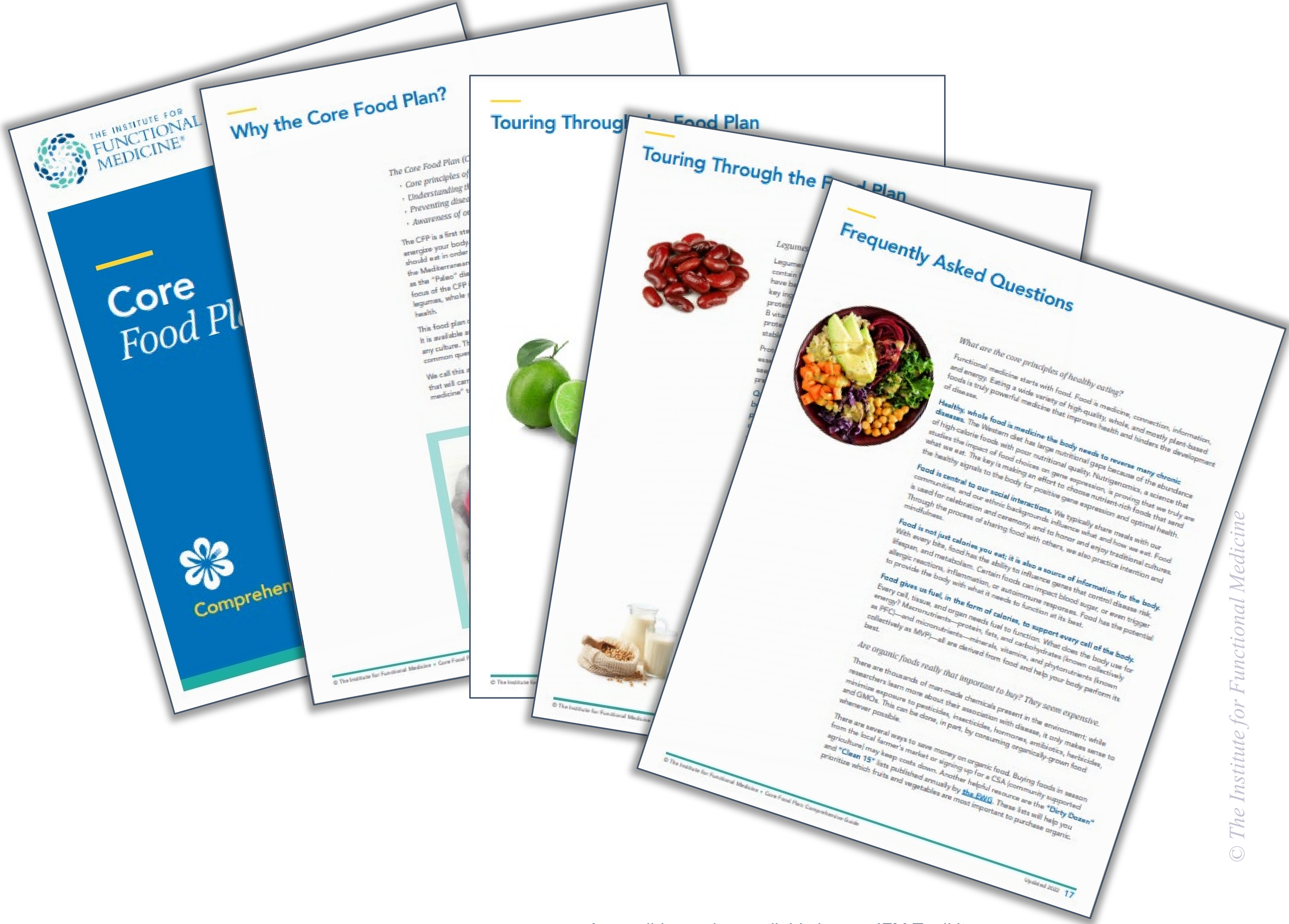
- The Core Food Plan and the Phytonutrient Spectrum are Toolkit Items. The Core Food Plan emphasizes how food is medicine, energy, information, and connection. Its features include: focus on whole foods, promote clean and organic foods, adequate quality protein, balanced quality fats, high in fiber, low in simple sugars, phytonutrient diversity, foundational eating plan

Modifications for Dietary Preferences and Needs

Common Dietary Preferences	• Vegetarian • Vegan • Pescatarian
Common Exclusions for Allergies or Sensitivities	• Gluten (or wheat) • Grains • Dairy products • Eggs • Nuts • Shellfish
Other Lifestyle Factors and Health Goals to Consider	• Weight loss • Weight gain • Enhanced athletic performance • Pregnancy and/or breastfeeding • Targeted health outcomes (controlling blood pressure, blood sugars, lipids)

Core Food Plan

Comprehensive Guide Available in Your Toolkit



Accessible version available in your IFM Toolkit.

Long Description: The Core Food Plan

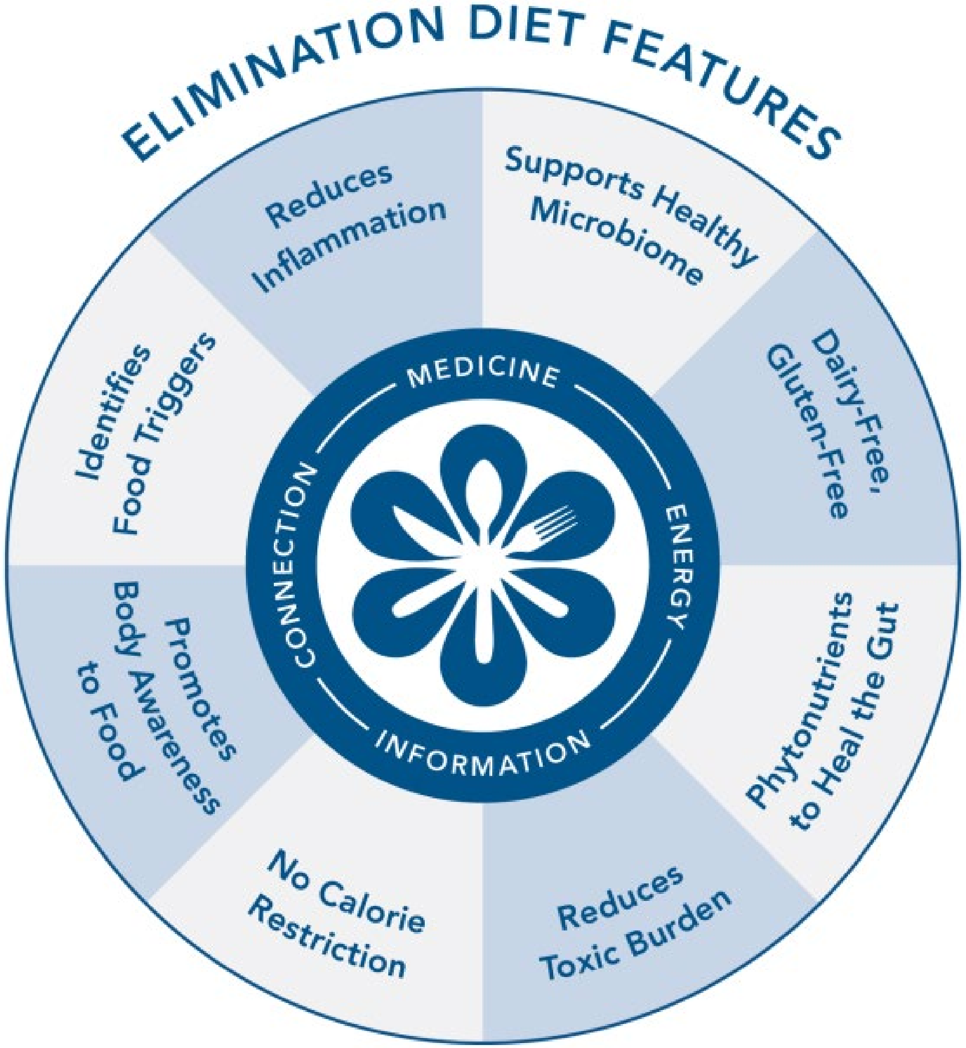
- The toolkit contains a Core Food Plan comprehensive guide

Next Steps

As You Continue Your Training at the APM Level

Next Steps
As You Continue Your Training at the APM Level

Elimination Diet and Its Variations



Long Description: Next Steps

- The Elimination Diet emphasizes food as medicine, energy, information, and connection. Its features include: Identifies Food Triggers, Reduces Inflammation, Supports Healthy Microbiome, Dairy-Free, Gluten-Free, Phytonutrients to Heal the Gut, Reduces Toxic Burden, No Calorie Restriction, Promotes Body Awareness to Food

Clinical Focus

- **Rebalance** interventions for DIGIN dysfunctions.

Therapeutic Interventions for Gut Dysfunction: The 5R Approach: <u>**Rebalance**</u>

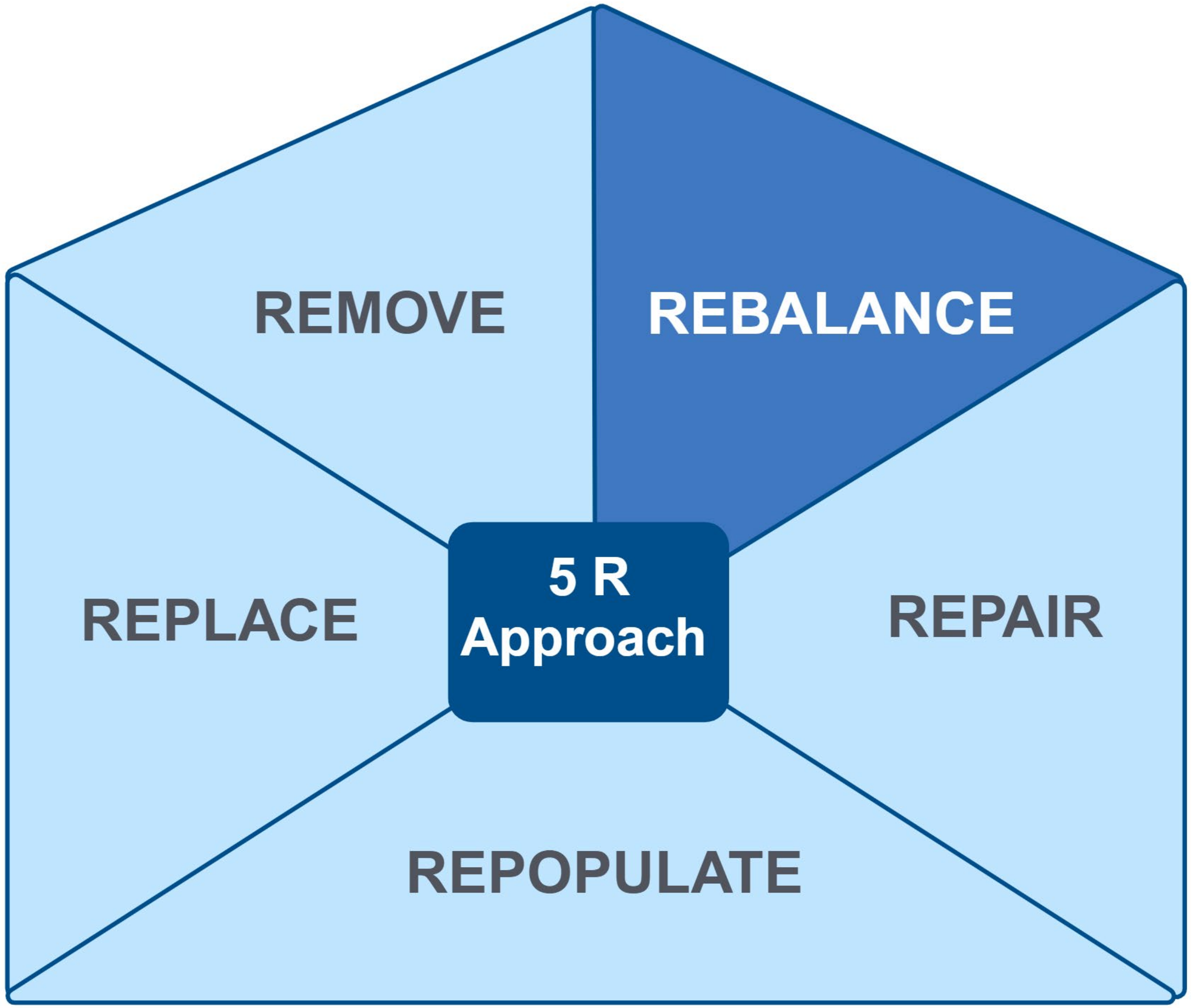
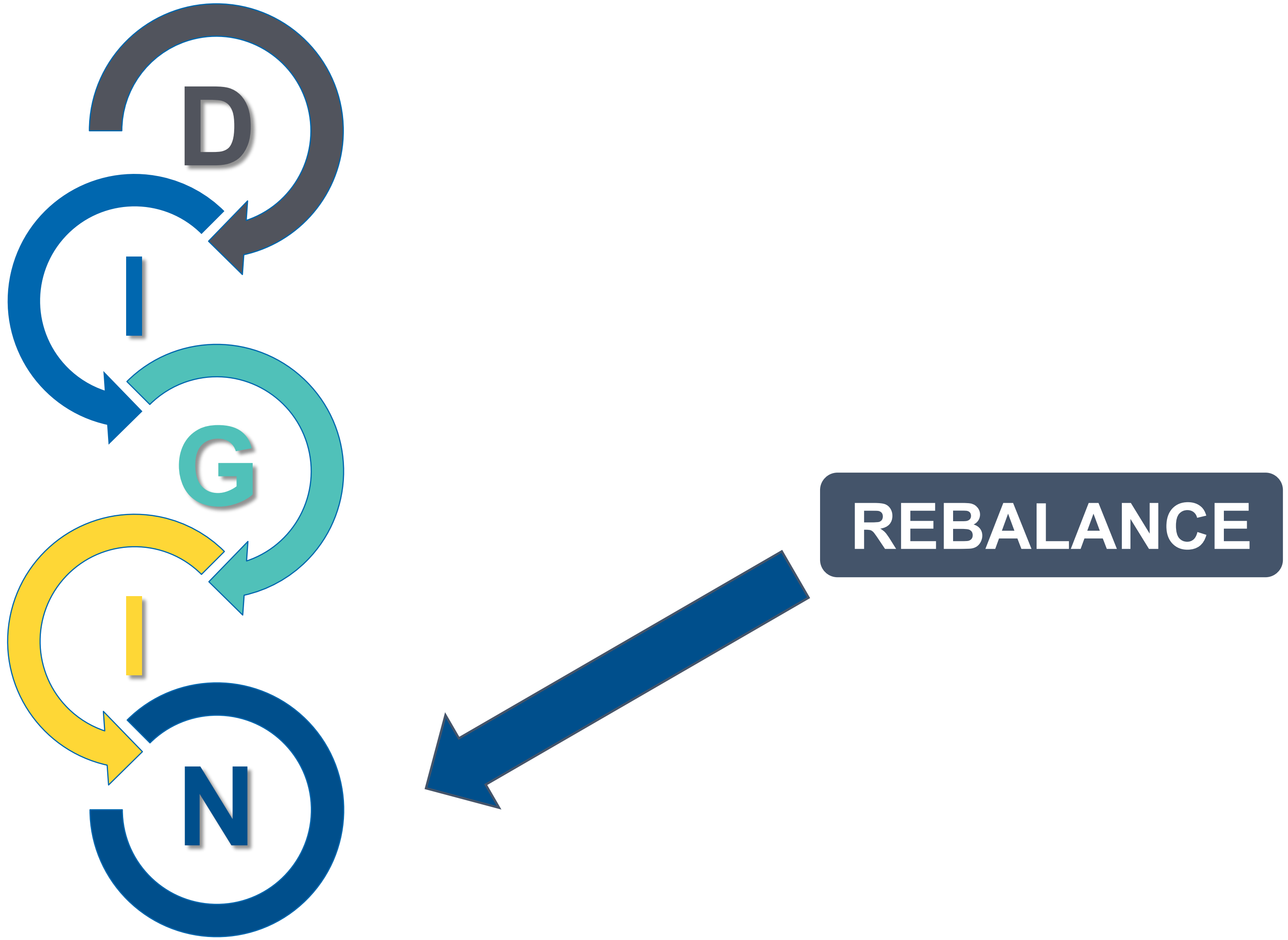


Image: Created with BioRender.com by The Institute for Functional Medicine

Long Description: Therapeutic Interventions for Gut Dysfunction: the 5 R Approach: Rebalance

- The 5 R Approach includes remove, replace, repopulate, repair, rebalance. Here the emphasis is on rebalance.

The **Rebalance** Interventions Are Most Relevant for Addressing ENS Dysfunction

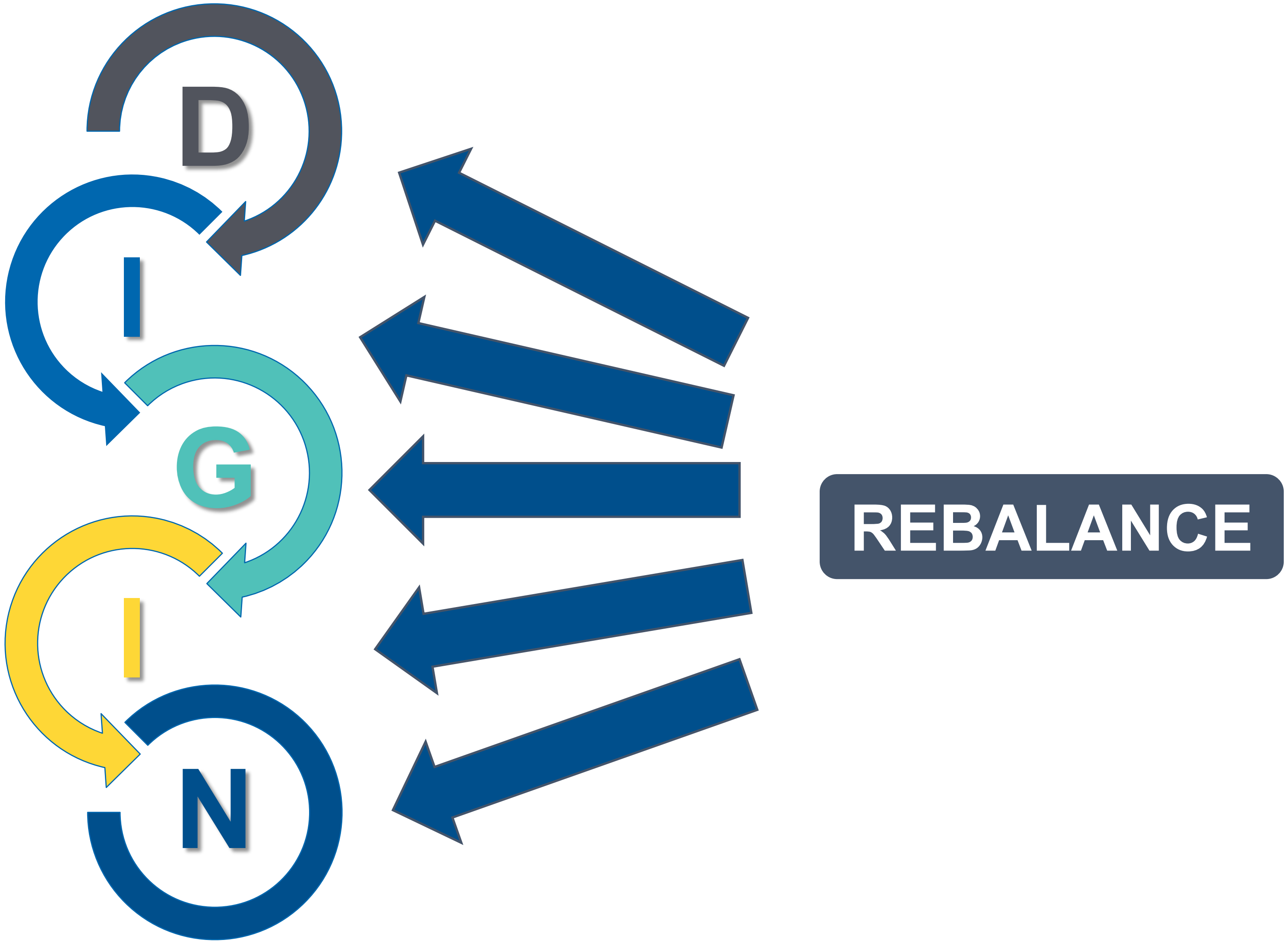


The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Long Description: The **Rebalance** Interventions Are Most Relevant for Addressing ENS Dysfunction

- Rebalance is most associated with N of the D I G I N approach.

There Are **Rebalance** Interventions That May Improve All Five DIGIN Dysfunctions



The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Long Description: There Are **Rebalance** Interventions That May Improve All Five D I G I N Dysfunctions

- Rebalance can address each node of D I G I N.

Rebalance: Foundational Interventions
(Must Be Personalized – Refer as Necessary)

Rebalance interventions support restorative processes in a patient's physiology.

Examples of Common Lifestyle, Mental, Emotional, and Spiritual Mediators of Gut Dysfunction

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships
• Adverse sleep behaviors	• Over exercise • Sedentarism	• Adverse meal behaviors	• Avoidable sources of stress	• Isolation, loneliness

See: "References: Rebalance"

Long Description: <u>**Rebalance**</u>: Foundational Interventions

- Common examples of modifiable lifestyle factors.
- Sleep: adverse sleep behaviors
- Physical Activity: over exercise; sedentarism
- Nutrition: adverse meal behaviors
- Stressors: avoidable sources of stress
- Relationships: isolation, loneliness

Rebalance: Foundational Interventions
(Must Be Personalized – Refer as Necessary)

Rebalance refers to providing support for restorative processes in a patient's life.

Therapeutic Approaches Emphasize Diet/Lifestyle Changes

- 'Scheduling' relaxation
- Mindful eating and better choices
- Heart rate variability/biofeedback
- Yoga, meditation, prayer, breathing, or other centering practices
- Psychotherapy
- Neuroplasticity strategies ("retraining the brain")
- Vagus nerve strategies

See: "References: Rebalance"

References

REBALANCE

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Representative Example of **Rebalance** on DIGIN

Psychoemotional Impact on the Microbiome

- Stress suppresses Lactobacillus, Bifidobacteria, and sIgA
- Catecholamines stimulate growth of gram-negative organisms (Yersinia, Pseudomonas)
  - 45-50% of total body production of norepinephrine occurs in mesenteric organs
- Anger or fear increases *Bacteroides fragilis*

See: “References: Psychoemotional Impact on the Microbiome”

References

PSYCHOEMOTIONAL IMPACT ON THE MICROBIOME

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Clinical Pearl

Rapid Identification of **Rebalance** Interventions

WHICH OF YOUR REGULAR HABITS, BEHAVIORS, AND PASTIMES:

… have a negative effect on your health?

… have a positive effect on your health?

HOW WOULD YOU COMPLETE THESE STATEMENTS?

My health would be better if I spent less time _______________.

My health would be better if I spent more time _______________.

Usual Sequence of the 5R Program

Primacy of the Remove and Rebalance Interventions

- Remove
- Replace
- Repopulate
- Repair
- Rebalance

Remove & Rebalance interventions may be empirically initiated while waiting for the results of stool testing and other evaluative and/or therapeutic interventions.

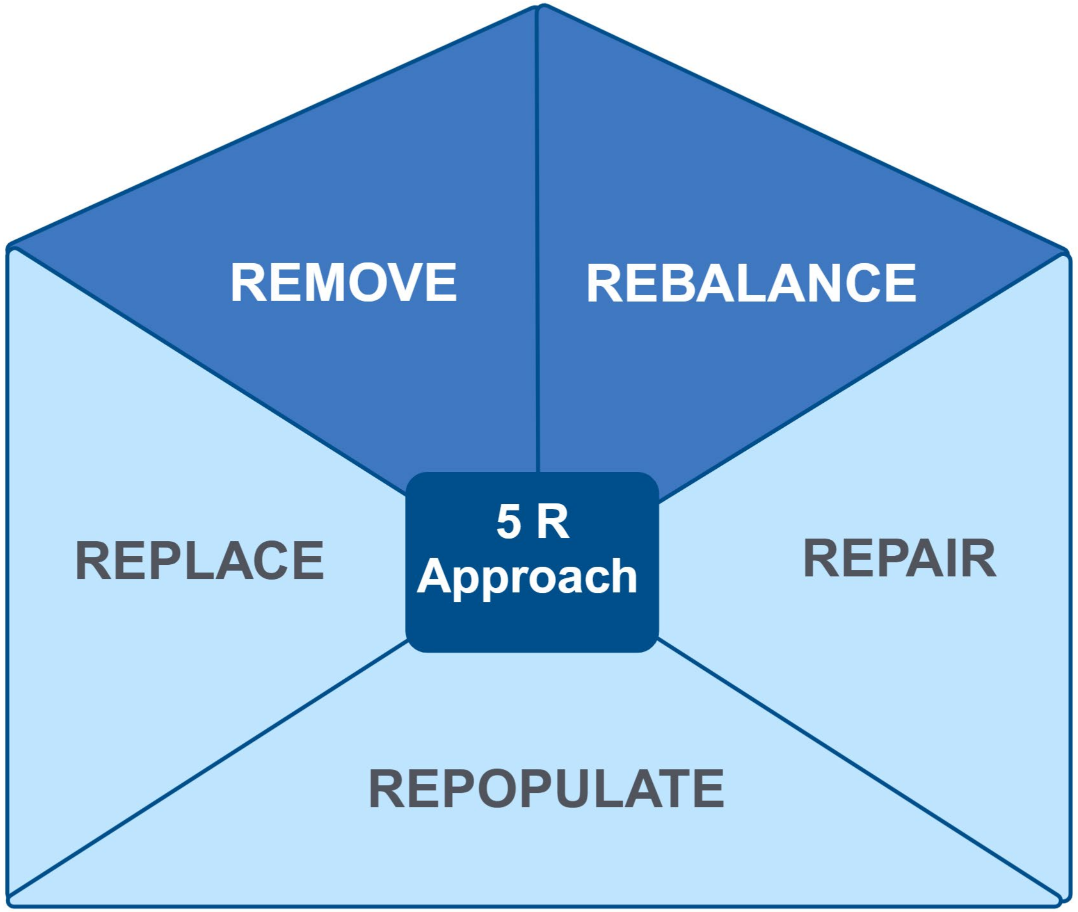


Image: Created with BioRender.com by The Institute for Functional Medicine

Long Description: Usual Sequence of the 5 R Program

The 5 R Approach includes remove, replace, repopulate, repair, rebalance. Here the emphasis is on Remove and Rebalance. Remove and Rebalance interventions may be empirically initiated while waiting for the results of stool testing and other evaluative and/or therapeutic interventions.

Clinical Focus

- **<u>Replace</u>** interventions for DIGIN dysfunctions.

Therapeutic Interventions for Gut Dysfunction: The 5R Approach: **Replace**

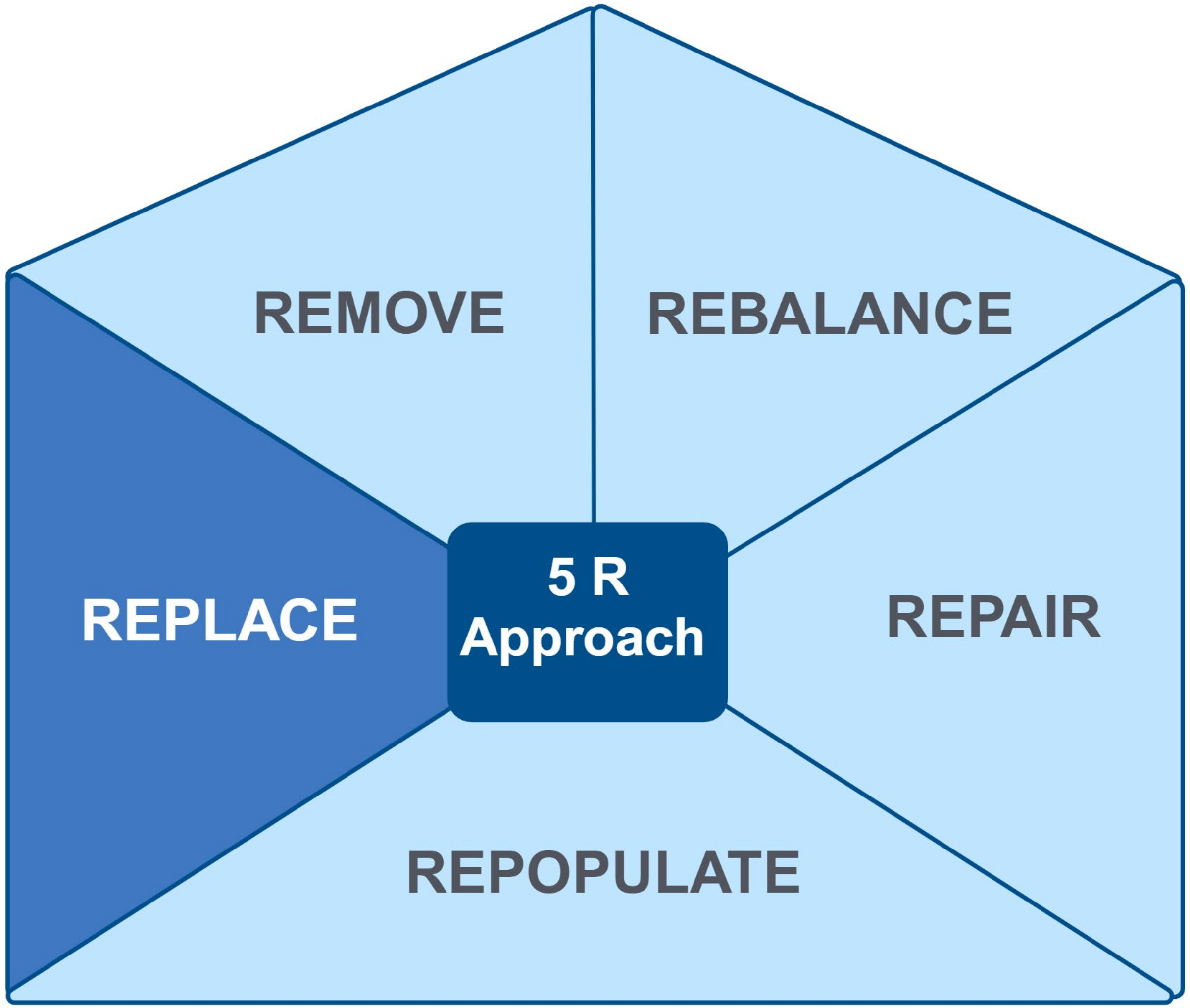


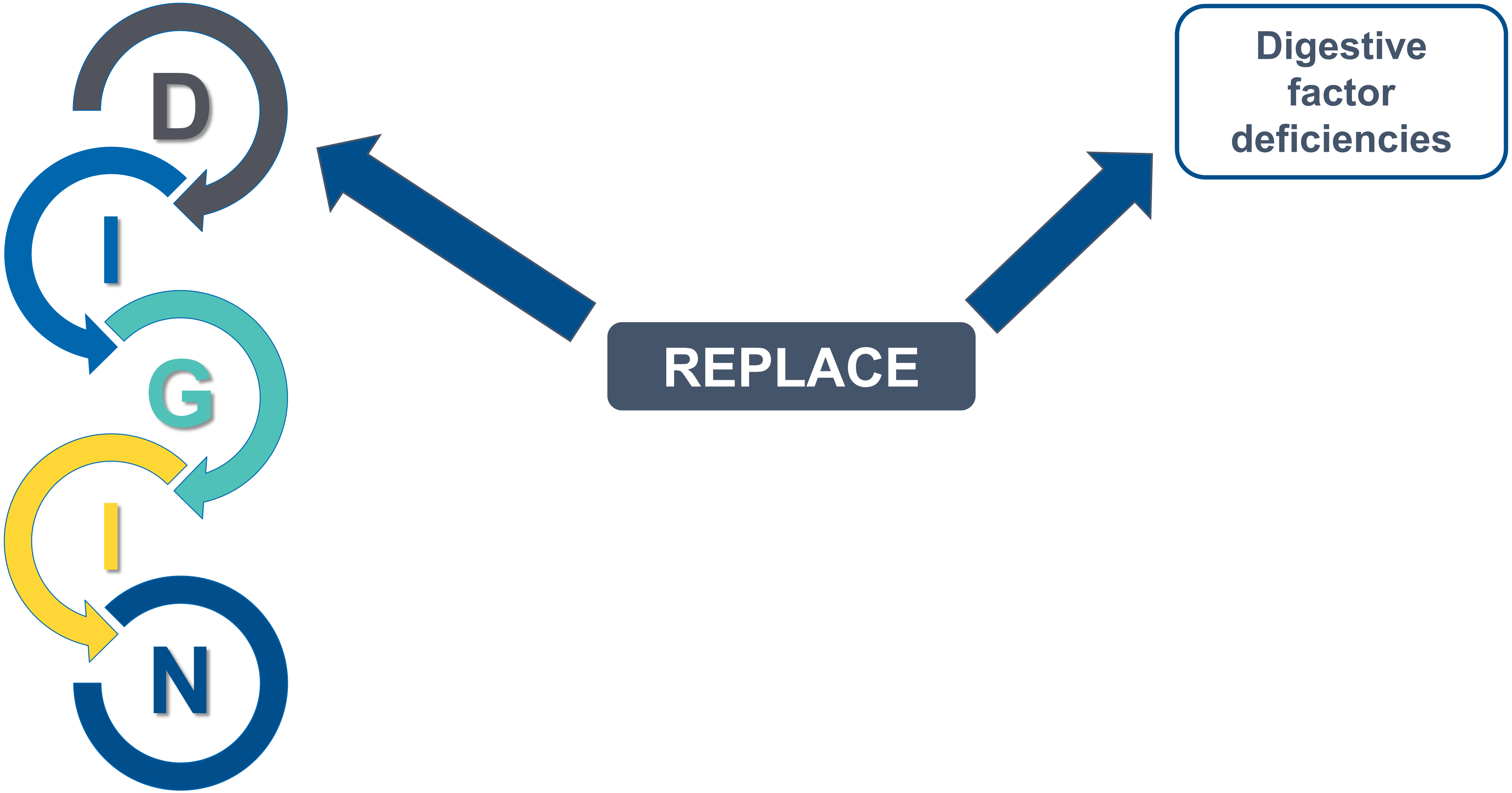
Image: Created with BioRender.com by The Institute for Functional Medicine

Long Description: Therapeutic Interventions for Gut Dysfunction

The 5 R Approach includes remove, replace, repopulate, repair, rebalance. Here the emphasis is on Replace.

Replace Interventions

Most Aligned With Maldigestion and Malabsorption



The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Long Description: Replace Interventions

For the D node of D I G I N, we think about replacing digestive factor deficiencies such as hydrochloric acid, pancreatic enzymes, and bile acids.

Digestive Deficiencies That May Need to Be **Replaced**

- Insufficient HCL
- Insufficient pancreatic enzymes
- Insufficient bile acids
- Insufficient intestinal brush border enzymes

The scope of this training will focus on insufficient HCl and pancreatic deficiencies

DIGIN Dysfunctions and Stool Test Biomarkers

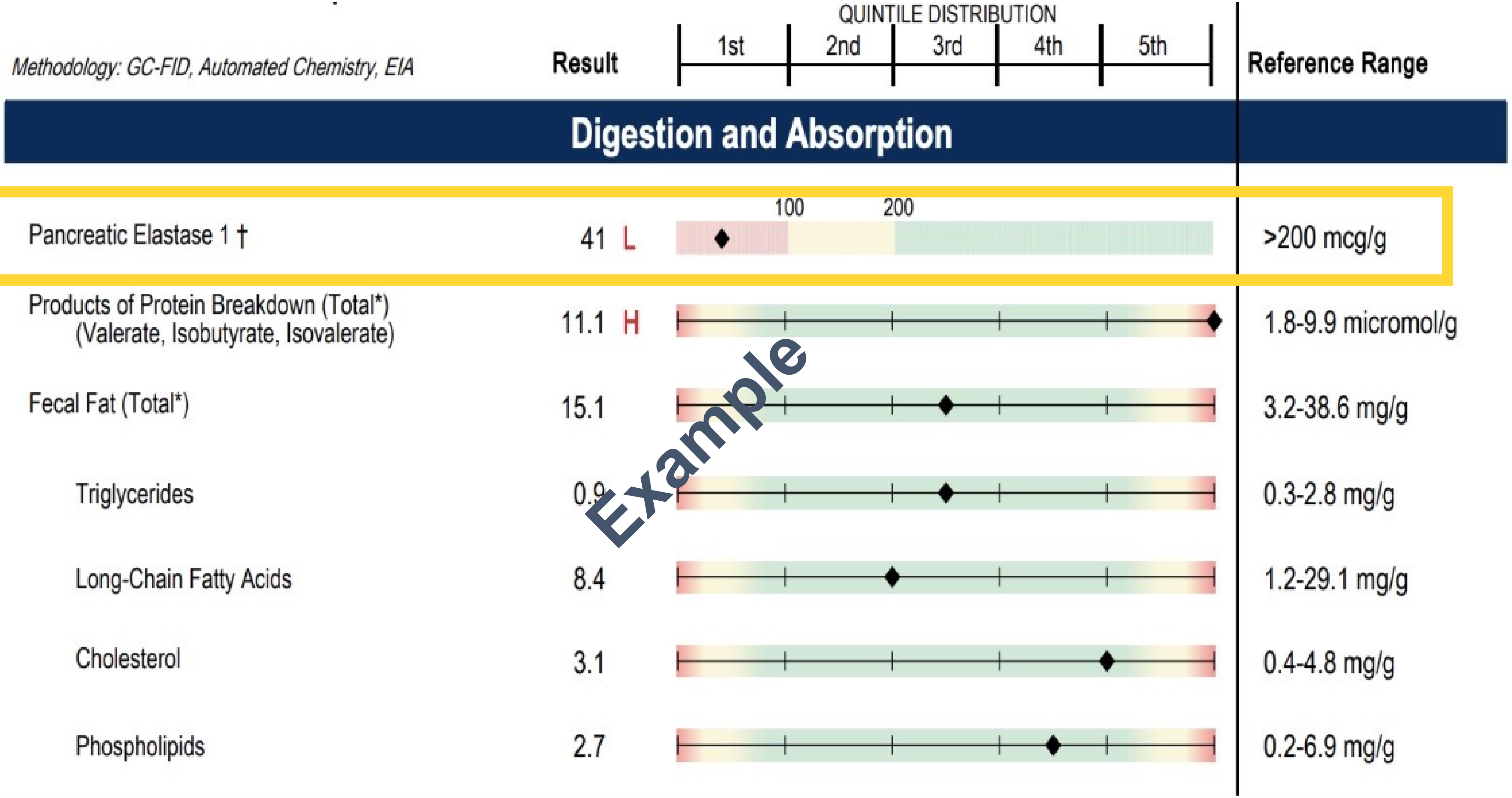
Gut DIGIN Dysfunction

- **Maldigestion/ Malabsorption**
  - Pancreatic Elastase
  - Valerate
  - Iso-Valerate
  - Iso-Butyrate
  - Fecal Fat
- **Intestinal Permeability**
  - No reliable stool test biomarker
- **Gut Dysbiosis**
  - Bacteria
  - Ova/Parasite
  - Yeast/Fungi
  - SCFA: Butyrate
  - Valerate
  - Propionate
  - Acetate
  - βGlucuronidase
- **Gut Inflammation**
  - Calprotectin
  - Lactoferrin
  - Eosinophil Protein X
  - sIgA
  - Lysozyme
- **ENS Dysfunction**
  - No reliable stool test biomarker

Long Description: D I G I N Dysfunctions and Stool Test Biomarkers

- Gut D I G I N Dysfunctions and Stool Test Biomarkers.
- Maldigestion and Malabsorption: pancreatic elastase, valerate, iso-valerate, iso-butyrate, fecal fat
- Intestinal Permeability: no reliable stool test biomarker
- Gut Dysbiosis: bacteria, Ova and parasite, yeast and fungi, S C F A – butyrate, valerate, propionate, acetate, beta-glucuronidase
- Gut Inflammation: calprotectin, lactoferrin, Eosinophil Protein X, secretory I G A, lysozyme
- E N S Dysfunction: no reliable stool test biomarker

Pancreatic Elastase



Methodology: GC-FID, Automated Chemistry, EIA

	Result		QUINTILE DISTRIBUTION: 1st, 2nd, 3rd, 4th, 5th	Reference Range
Digestion and Absorption				
Pancreatic Elastase 1 †	41	L	100, 200	>200 mcg/g
Products of Protein Breakdown (Total*) (Valerate, Isobutyrate, Isovalerate)	11.1	H		1.8-9.9 micromol/g
Fecal Fat (Total*)	15.1			3.2-38.6 mg/g
Triglycerides	0.9			0.3-2.8 mg/g
Long-Chain Fatty Acids	8.4			1.2-29.1 mg/g
Cholesterol	3.1			0.4-4.8 mg/g
Phospholipids	2.7			0.2-6.9 mg/g

Example

Image courtesy of Elizabeth Boham, MD, MS, RD

Long Description: Pancreatic Elastase

This is an example of a stool test result show very low pancreatic elastase and elevated products of protein breakdown including valerate, isobutyrate, and isovalerate. The levels of total fecal fat, triglycerides, long-chain fatty acids, and cholesterol are within the normal ranges. These results suggest maldigestion and malabsorption.

Replace

Foundational Interventions for Deficient Digestive Factors

Therapeutic Approaches Concurrent With Diet and Lifestyle May Include:

- Digestive Factor Repletion:
  - Gastric Acidity Support Factors
  - Pancreatic Enzymes
  - Fiber to Support Motility and General GI Function

Hypochlorhydria
Clinical Findings That May Warrant (Empirical) Treatment

- Meal hygiene behaviors
  - Recall **cephalic phase** of digestion
  - Stress/sympathetic tone
- Dyspepsia within one-hour postprandial
- +/- Stool test products of protein breakdown elevated
- Physical exam findings, especially related to protein deficiency

Foundational Interventions to Support Gastric Acidity

Best With a Protein-Containing Meal:

- Bitters
  - Examples: Ginger, wormwood, cinnamon, myrrh, orange peel, fennel, dandelion root, artichoke, rhubarb
- Vinegar
- Umeboshi plums
- Digestive enzymes with acid pH range

Lifestyle:

- Stress management
- Acupuncture

See: "References: Supporting Gastric Acidity"

References

SUPPORTING GASTRIC ACIDITY

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Pancreatic Exocrine Insufficiency
Clinical Findings That May Warrant (Empirical) Treatment

- Dyspepsia 2-4 hours postprandial
- Stool test = low pancreatic elastase
  - +/- elevated products of protein breakdown
  - +/- elevated fecal fat
- Physical exam indicators of nutrient deficiencies
- Medical conditions affecting the pancreas

Food-Derived Enzymes

- **Bromelain** (from pineapple)
  - Enzymatic breakdown of protein and is associated with reduced inflammation
- **Papain** (from papaya) – *Second Tier Intervention*
  - Dose: typically chewable: 100-250 mg

See: "References: Food-Derived Enzymes"

References: Food Derived Enzymes

Papain reduces gastric inflammatory pain and improves digestive symptoms (Moderate Evidence)

- Muss C, Mosgoeller W, Endler T. Papaya preparation (Caricol®) in digestive disorders. Neuro Endocrinol Lett. 2013;34(1):38-46.
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Bromelain

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Current FDA-Approved “PERTs”

Pancreatic Enzyme Replacement Therapies

Second-Tier Therapeutics

Brand	Units of lipase (USP)
Creon	3,000; 6,000; 12,000; 24,000; 36,000
Zenpep	3,000; 5,000; 10,000; 15,000; 20,000; 25,000
Pancreaze	4,200; 10,500; 16,800; 21,000
Ultresa	13,800; 20,700; 23,000
Viokase	10,440; 20,880 (requires acid suppression)
Pertzye	8,000; 16,000

- Adapted from: Struyvenberg MR, Martin CR, Freedman SD. Practical guide to exocrine pancreatic insufficiency – breaking the myths. *BMC Med*. 2017;15(1):29. doi:10.1186/s12916-017-0783-y

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Pancreatin: Animal-Derived

TAKE WITH FOOD

Second-Tier therapeutics

Pancreatin (serving size 3 tablets)

Protease 156,000 USP
Amylase 156,000 USP
Lipase 24,960 USP

Pancreatin 8X 125 mg (1 tablet)

Amylase 25,000 USP units
Protease 25,000 USP units
Lipase 2,000 USP units

Pancreatin Range: 500-2,500 units/kg/meal

Pancreatin 4X 500 mg (1 tablet)

Amylase 50,000 USP units
Protease 50,000 USP units
Lipase 4,000 USP units

See: "References: Pancreatin: Animal-Derived"

References

PANCREATIN: ANIMAL-DERIVED

Pancreatic enzyme replacement reduces digestive symptoms in patients with exocrine pancreatic insufficiency (Strong)

- de la Iglesia-García D, Huang W, Szatmary P, et al. Efficacy of pancreatic enzyme replacement therapy in chronic pancreatitis: systematic review and meta-analysis. Gut. 2017;66(8):1354-1355. doi:10.1136/gutjnl-2016-312529
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Foods That Act as Cholagogues
When Insufficient Bile Is Clinically Suspected

- Coffee
- Radishes (horse, red, daikon)
- Dandelion
- Chicory and other bitter greens
- Artichoke

- Pizzorno JE, Murray MT. *Textbook of Natural Medicine*. Elsevier/Churchill Livingstone; 2013.
- See: “References: Foods That Act as Cholagogues”

References

FOODS THAT ACT AS CHOLAGOGUES

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  - Zhang YP, Li WQ, Sun YL, Zhu RT, Wang WJ. Systematic review with meta-analysis: coffee consumption and the risk of gallstone disease. *Aliment Pharmacol Ther*. 2015;42(6):637-648. doi:10.1111/apt.13328
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REPLACE = REPLETE
Nutrient Deficiencies

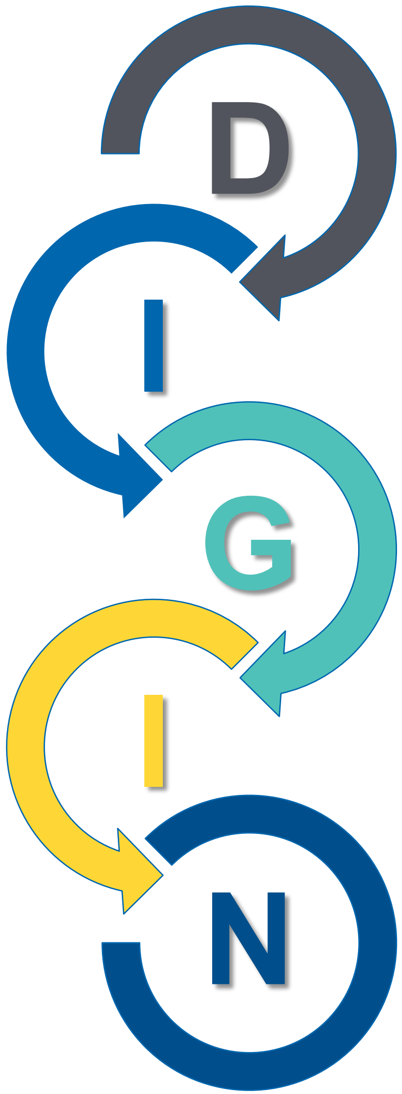
Maldigestion from any cause may be a mediator of nutrient deficiencies.

Foundational Interventions:

Core Food Plan

Phytonutrient Spectrum

Shifting the Focus to Malabsorption



***D**igestion/Absorption*

***I**ntestinal Permeability*

***G**ut Microbiota/Dysbiosis*

***I**nflammation/Immune*

***N**ervous System*

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Long Description: Shifting the Focus to Malabsorption

- The D I G I N Approach
- D: Digestion/Absorption
- I: Intestinal Permeability
- G: Gut Microbiota/Dysbiosis
- I: Inflammation/Immune
- N: Nervous System

REPLACE = REPLETE
Nutrient Deficiencies

Malabsorption or…

Maldigestion from any cause may be a mediator of nutrient deficiencies.

<u>Foundational Interventions</u>:

Core Food Plan

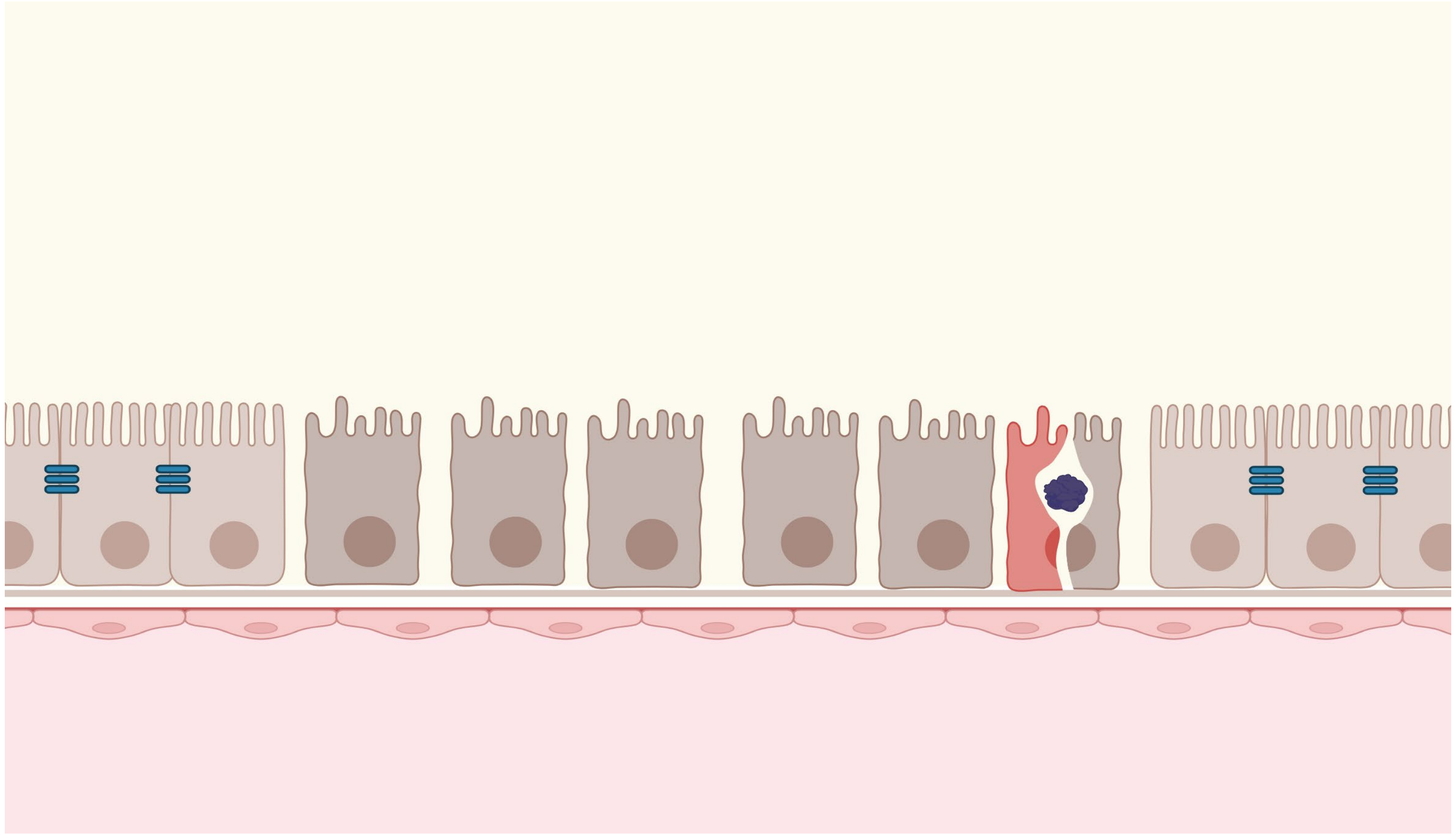
Phytonutrient Spectrum

Long Description: Usual Sequence of the 5 R Program

The 5 R Approach includes remove, rebalance, repair, repopulate, and replace. Here the emphasis is on rebalance. Remove and Rebalance interventions may be empirically initiated while waiting for the results of stool testing and other evaluative and/or therapeutic interventions.

Recall how damage to the intestinal lining can lead to malabsorption

Malabsorption ⟷ Repair



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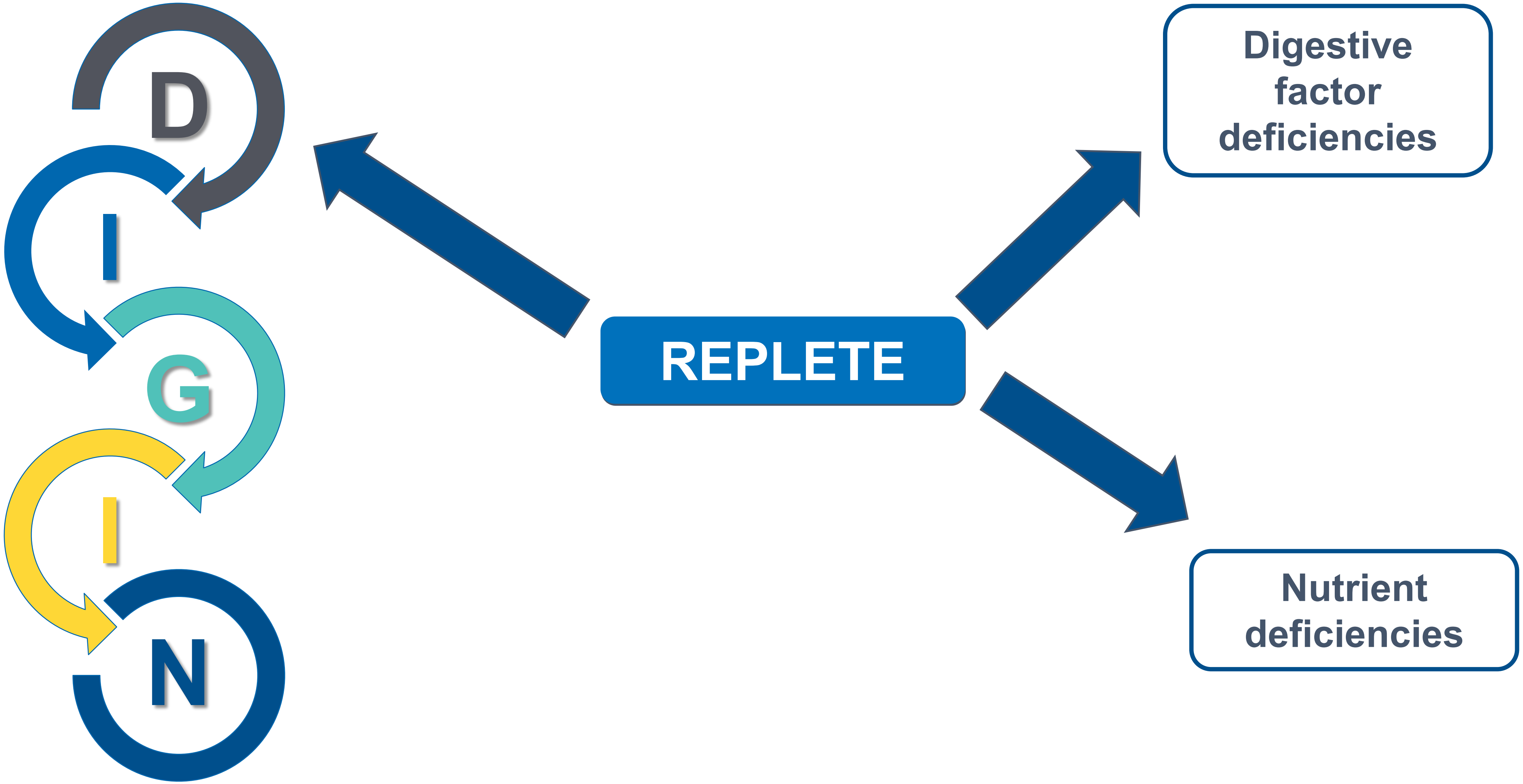
Long Description: Malabsorption and Repair

- Damage to the intestinal lining can cause malabsorption. Replacing factors that improve digestion will help with repair.

Replace & Replete Interventions

Digestive Factor Support AND Repletion of Nutrient Deficiencies



The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Long Description: Replace and Replete Interventions

- D I G I N with an emphasis on Replate. We replate digestive factor deficiencies and nutrient deficiencies.

Mediators of Maldigestion/Malabsorption

Rationale for Considering All 5 of the 5Rs

- **R**emove ← Meds, toxins, problem foods damage intestine
- **R**eplace ← Deficiency of stomach, pancreas, gallbladder brush border digestive factors
- **R**epopulate ← Dysbiotic microbes consume nutrients and damage intestinal tissue
- **R**epair ← Brush border injury reduces secretion of CCK, secretin, and disaccharidases
- **R**ebalance ← Stress impairs secretion of HCl, digestive enzymes

Long Description: Mediators of Maldigestion and Malabsorption

Mediators of Maldigestion and Malabsorption for all five of the five R's.
Remove: medications, toxins, problem foods that are damaging the intestine
Replace: deficiency of stomach, pancreas, gallbladder, brush border digestive factors
Repopulate: dysbiotic microbes consume nutrients and damage intestinal tissue
Repair: brush border injury reduces secretion of CCK, secretin, and disaccharidases
Rebalance: stress impairs secretion of HCl, digestive enzymes

Therapeutic Interventions for Gut Dysfunction: Part 2

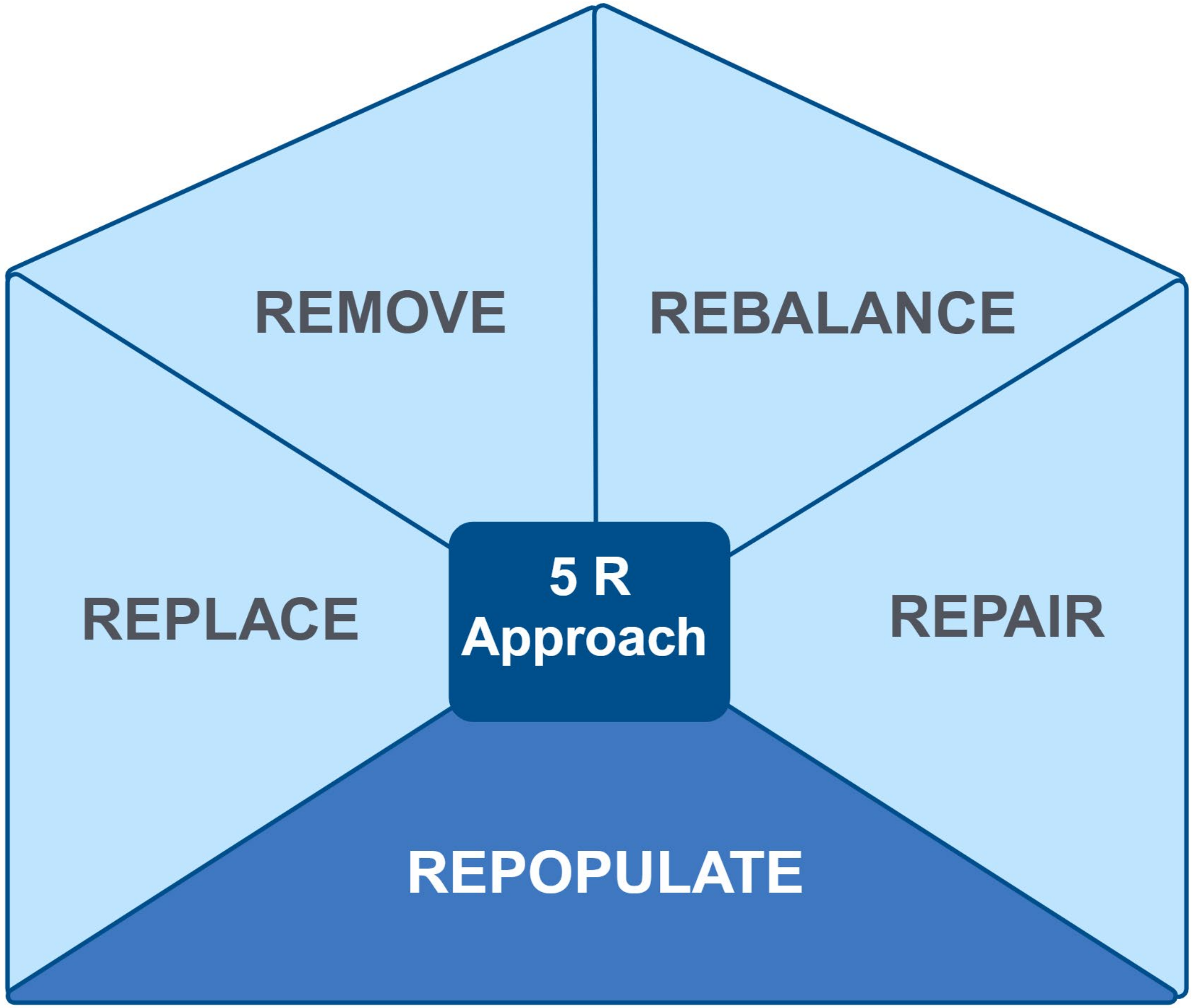
The 5R Approach

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- **<u>Repopulate</u>** interventions for DIGIN dysfunctions.

Therapeutic Interventions for Gut Dysfunction: The 5R Approach: **Repopulate**



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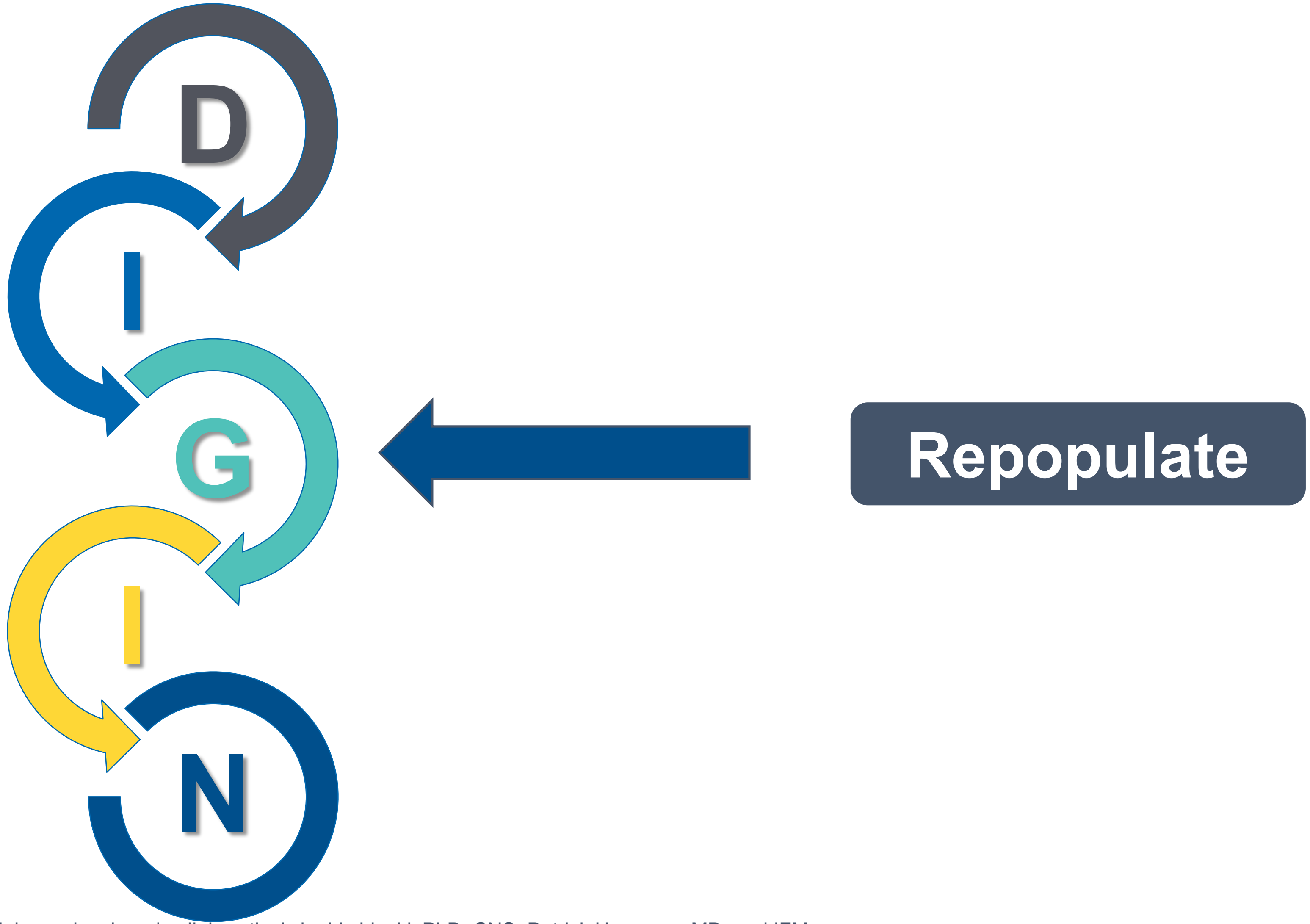
Long Description: The 5 R Approach

The 5 R Approach includes remove, replace, repopulate, repair, rebalance. Here the emphasis is on Repopulate.

Repopulate Interventions Are Most Relevant for Addressing Gut Dysbiosis



The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Long Description: Repopulate

- Repopulate most closely aligns with G on D I G I N or the gut microbiome.

Dysbiosis is difficult to diagnose by clinical findings alone.

Laboratory stool testing is often warranted to confirm diagnosis, recognize patterns, and guide therapeutic regimen.

Repopulate by **Feeding** and **Seeding**

Prebiotics

Probiotics

DIGIN Dysfunctions and Stool Test Biomarkers

Gut DIGIN Dysfunction

- **Maldigestion/ Malabsorption**
  - Pancreatic Elastase
  - Valerate
  - Iso-Valerate
  - Iso-Butyrate
  - Fecal Fat
- **Intestinal Permeability**
  - No reliable stool test biomarker
- **Gut Dysbiosis**
  - Bacteria
  - Ova/Parasite
  - Yeast/Fungi
  - SCFA: Butyrate
  - Valerate
  - Propionate
  - Acetate
  - βGlucuronidase
- **Gut Inflammation**
  - Calprotectin
  - Lactoferrin
  - Eosinophil Protein X
  - sIgA
  - Lysozyme
- **ENS Dysfunction**
  - No reliable stool test biomarker

Long Description: D I G I N Dysfunctions and Stool Test Biomarkers

- Gut D I G I N Dysfunctions and Stool Test Biomarkers.
- Maldigestion and Malabsorption: pancreatic elastase, valerate, iso-valerate, iso-butyrate, fecal fat
- Intestinal Permeability: no reliable stool test biomarker
- Gut Dysbiosis: bacteria, Ova and parasite, yeast and fungi, S C F A – butyrate, valerate, propionate, acetate, beta-glucuronidase
- Gut Inflammation: calprotectin, lactoferrin, Eosinophil Protein X, secretory I G A, lysozyme
- E N S Dysfunction: no reliable stool test biomarker
- Here we are emphasizing the gut dysbiosis node: the short-chain fatty acids: butyrate, valerate, propionate, and acetate.

Stool Test Biomarkers

Gut Microbiome Metabolites			
Metabolic			
Short-Chain Fatty Acids (SCFA) (Total*) (Acetate, n-Butyrate, Propionate)	16.5	L	>=23.3 micromol/g
n-Butyrate Concentration	2.2	L	>=3.6 micromol/g
n-Butyrate %	13.3		11.8-33.3 %
Acetate %	60.5		48.1-69.2 %
Propionate %	26.1		<=29.3 %
Beta-glucuronidase	758		368-6,266 U/g

Lab image courtesy of Elizabeth Boham, MD, MS, RD

Long Description: Stool Test Biomarkers

This is a sample stool test show low levels of short chain fatty acids. In this example, total short-chain fatty acids including acetate, N Butyrate, and propionate are low. The N Butyrate concentration and N Butyrate percentage are also low. Propionate percentage is elevated.

Feeding: Stool Test Indicators

Laboratory stool assessment indicating:

- Low SCFAs and/or low butyrate
- A deficiency in the commensal flora

Therapeutic considerations:

1) Dietary recommendations incorporating prebiotics
2) Supplemental recommendations incorporating prebiotics

Feeding the Microbiome: Dietary Fiber Intake

- Paleolithic (estimated) intake **100-125 g/day**
- Recommended daily intake **25-35 g/day**
- Actual (average) intake in US **8-15 g/day**
- **Physiologic Properties**
  1. Slows transit in small bowel
  2. Increases stool bulk
  3. Holds on to water
  4. Increases estrogen excretion
  5. Binds minerals and organotoxins
  6. Stimulates bacterial growth
  7. Metabolized to SCFA

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Eating for Gut Health

A healthy gut lays the foundation for a healthy mind and body, and supporting it starts with the food on your plate. While many factors influence gut health, the three main food-related ones are when you eat, how you eat, and what you eat. This guide provides practical strategies to help you nurture digestion and a thriving gut. Think of eating for gut health as a long-term lifestyle change, not a quick fix – small steps can lead to big improvements, so start with a few changes and build from there.

Understanding the Connection Between Digestion, Gut Health, and Whole-Body Health

"Gut health" refers to the well-being of your **digestive system**, a group of organs that breaks down and absorbs food and nutrients, and your **gut microbiota**, the microorganisms that live in it. A healthy gut is connected to a reduced risk of chronic diseases, ensures you get the nutrition you need, protects your body from harmful substances, and regulates inflammation.[1,2] When your gut is out of balance, it is hard to feel good and function at your best.

Digestion, the process of breaking down food in the digestive system, affects gut health. It begins in your brain and ends with waste elimination through a bowel movement. Understanding each stage of digestion can help you make changes to your food choices and eating behaviors to support optimal gut health.

Image courtesy of 123RF

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Accessible version available in your toolkit.

Long Description: Feeding the Microbiome

- Toolkit Item: Eating for Gut Health

Feeding: Therapeutic Considerations

Foundational Prebiotic-Rich Foods

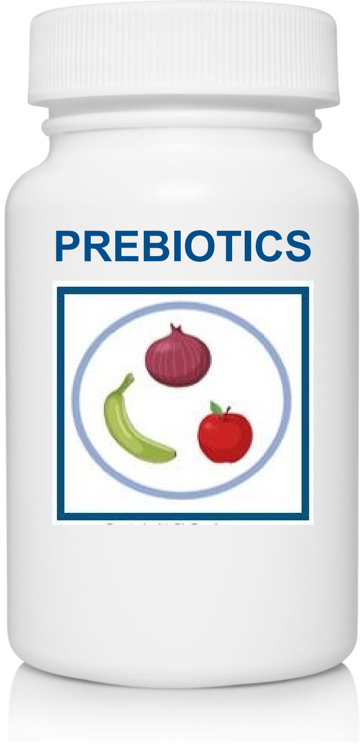
- Artichoke
- Asparagus
- Avocado
- Bananas (under ripe)
- Barley
- Beet root
- Bran
- Burdock root
- Chia seeds
- Chicory
- Chinese chives
- Cocoa
- Dandelion greens
- Eggplant
- Dairy products (CLA), yogurt, cottage cheese
- Flaxseeds
- Fruit
- Garlic
- Green tea
- Honey
- Jerusalem artichokes
- Jicama
- Kefir
- Leeks
- Legumes
- Lentils
- Onions
- Peas
- Radishes
- Root vegetables
- Plantains
- Potatoes
- Rye
- Sea vegetables
- Soybeans
- Spices and herbs
- Sweet potatoes
- Vegetables
- Yams

Feeding the Microbiome: Prebiotics

Second-Tier Intervention: Supplemental Prebiotics

Treatment time – indefinite

- **FOS/fructo-oligosaccharides:** 1,000-5,000 mg, QD-TID



- See: "References: Prebiotics"
- Image: Created with BioRender.com by The Institute for Functional Medicine

References

PREBIOTICS

Fructo-oligosaccharide (FOS) supplementation improves symptoms of constipation (Moderate, Conflicting Evidence)

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PREBIOTICS

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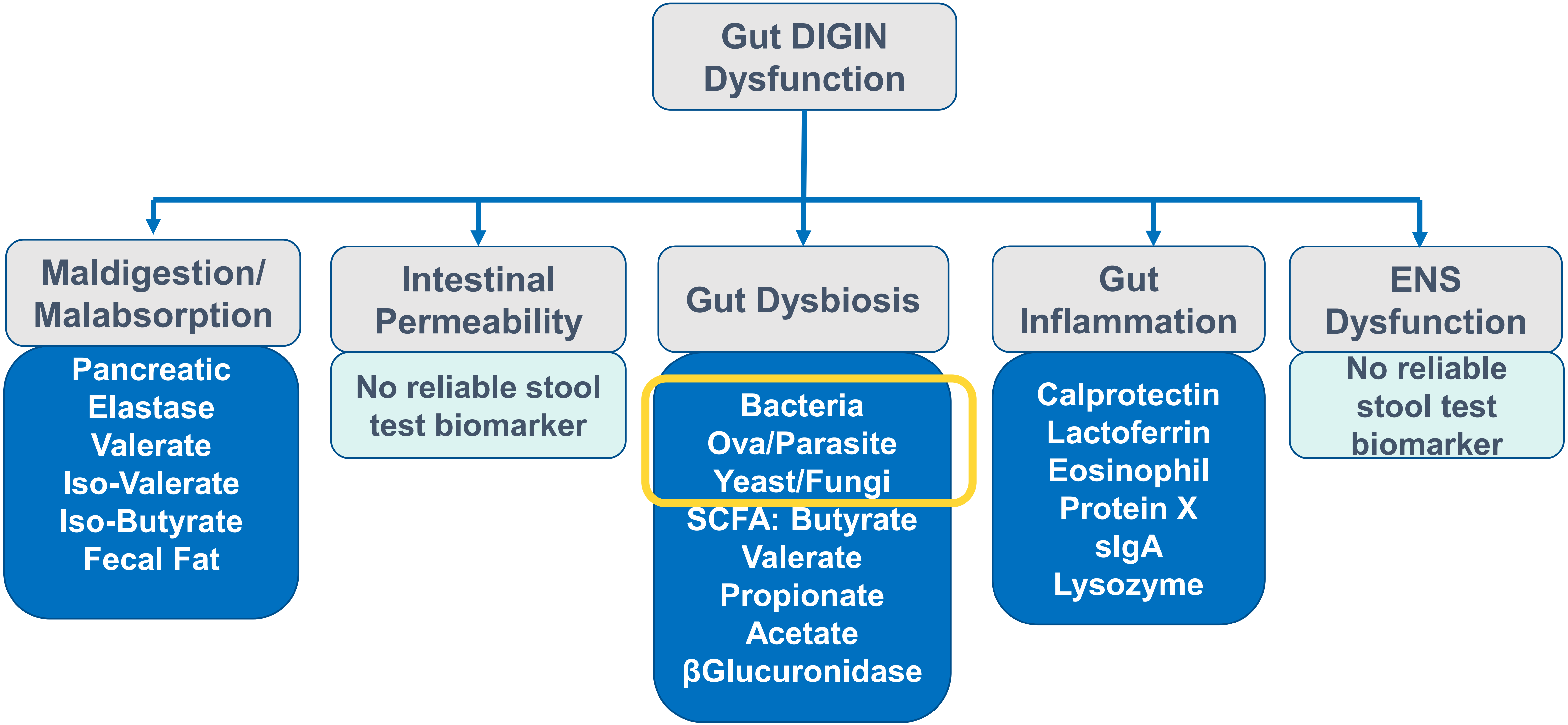
DIGIN Dysfunctions and Stool Test Biomarkers



Long Description: D I G I N Dysfunctions and Stool Test Biomarkers

- Gut D I G I N Dysfunctions and Stool Test Biomarkers.
- Maldigestion and Malabsorption: pancreatic elastase, valerate, iso-valerate, iso-butyrate, fecal fat
- Intestinal Permeability: no reliable stool test biomarker
- Gut Dysbiosis: bacteria, Ova and parasite, yeast and fungi, S C F A – butyrate, valerate, propionate, acetate, beta-glucuronidase
- Gut Inflammation: calprotectin, lactoferrin, Eosinophil Protein X, secretory I G A, lysozyme
- E N S Dysfunction: no reliable stool test biomarker
- Here we are emphasizing gut dysbiosis node: bacteria, ova and parasite, yeast, and fungi.

Treating Pathogenic Organisms vs Treating Dysbiosis

- For dysbiosis, consider a continuum for affecting change in the gastrointestinal milieu:
  - **Dietary changes**
  - **Pre- and probiotics**

Dysbiosis
Non-pathogens

  - **Anti-microbial botanicals***
  - **Anti-microbial medications***

Dysbiosis/Infection
Pathogens

- Anti-microbial pharmaceutical and/or botanical treatments should be reserved for significant and persisting dysbiosis and for patients with true pathogenic infections.*

* indicates, third-tier therapeutics to be discussed in Advanced Practice Modules

<u>**Seeding**</u>: Stool Test Indicators

Laboratory stool assessment indicating:

- A deficiency in the commensal flora
- Elevated potentially pathogenic bacteria

Therapeutic considerations:

1) Dietary recommendations incorporating fermented foods/probiotic-rich foods
2) Supplemental recommendations incorporating probiotics

Seeding: Therapeutic Considerations

Foundational Probiotic-Rich Foods

- Yogurt/kefir
- Miso
- Natto
- Tempeh
- Sauerkraut
- Kimchi
- Raw pickles
- Fermented anything
- Root and ginger beers
- Sourdough, essene bread
- Fermented vegetables/sausages
- Pulke
- Kombucha
- Buttermilk
- Raw whey
- Raw vinegars

Seeding: Therapeutic Considerations

Second-Tier Intervention: Supplemental Probiotics

- Which organism to use?
- Which product?
- For what conditions?
- What dose?
- For how long?
- Any side effects to be aware of?
- How much does it cost?

Practical Issues:

- Typically $1 to $3 per day
  - VSL#3®: $66 for a 30-day supply
  - Culturelle® (LGG): $18 for 30-day supply
  - Custom Probiotics CP-1: $60 for 30-day supply

- May need several months of therapy to see an effect
- Likely stop working after discontinued
- Concentration (dose) highly variable

See: "References: Probiotics"

References

PROBIOTICS

Probiotic supplementation at a dose of 10^9 to 10^10 CFU daily may have large beneficial effects on abdominal pain and global symptoms. (Strong)

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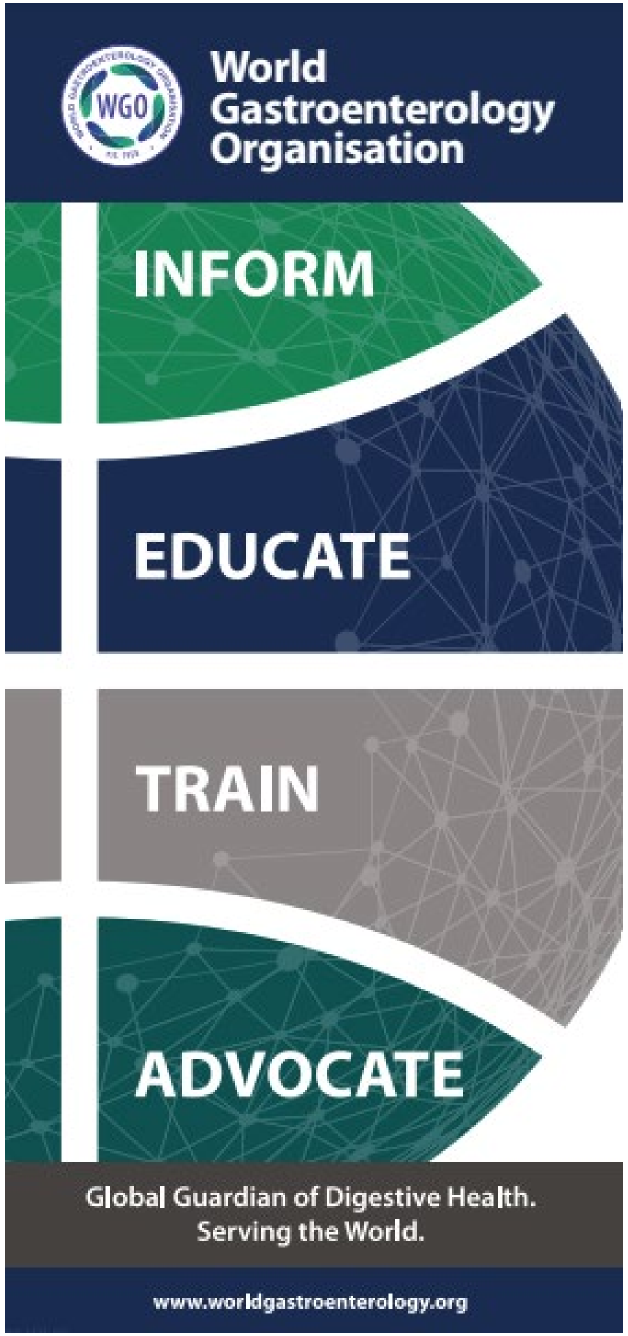
Seeding the Microbiome
Probiotic Prescribing: Resources to Consider

- Natural Medicines Database
  - Free with IFM Membership
- USProbioticguide.com
  - Free
- Probiotic Advisor
  - Subscription

<u>**Seeding**</u> the Microbiome Probiotic Prescribing:
Resources to Consider

- World Gastroenterology Organisation Global Guidelines
- *https://www.worldgastroenterology.org/guidelines/probiotics-and-prebiotics*



Feeding and Seeding the Microbiome

- **Dietary Prebiotics** (Food as Medicine)
  - Prebiotics
  - Fermented foods
  - Soluble fiber
  - Synbiotics = Pre + Probiotics

- **Supplemental Probiotics** (Bugs as Drugs)
  - Dosage?
  - Strains?
  - Safety?

Repopulate: Interventions

(For non-pathogen dysbiosis and/or low SCFA identified on stool testing)

Repopulate refers to the reintroduction of desirable GI microflora (prebiotics, probiotics, synbiotics) to obtain a more desirable balance to the intestinal milieu.

Therapeutic Approaches Beyond Diet/Lifestyle May Include:

Probiotics (common agents):

- Bifidobacteria strains
- Lactobacillus strains
- Saccharomyces boulardii

Prebiotics (common agents):

- Inulin or fructo-oligosaccharides (FOS)
- Various other soluble fibers

Synbiotics (common agents):

- Bifidobacteria and FOS
- Lactobacillus and inulin

Resources Available in Your Toolkit

Accessible version available in your toolkit.

Long Description: Resources Available in your Toolkit

There are several toolkit handouts on these topics including:
What is Fiber?
Food Sources of Dietary Fiber
Probiotic and Prebiotic Foods

Next Steps

As You Continue Your Training at the APM Level

Next Steps

As You Continue Your Training at the APM Level

Botanical and pharmaceutical
anti-microbial therapeutic regimens:
Weeding

Clinical Focus

- **<u>Repair</u>** interventions for DIGIN dysfunctions.

Therapeutic Interventions for Gut Dysfunction: The 5R Approach: **Repair**

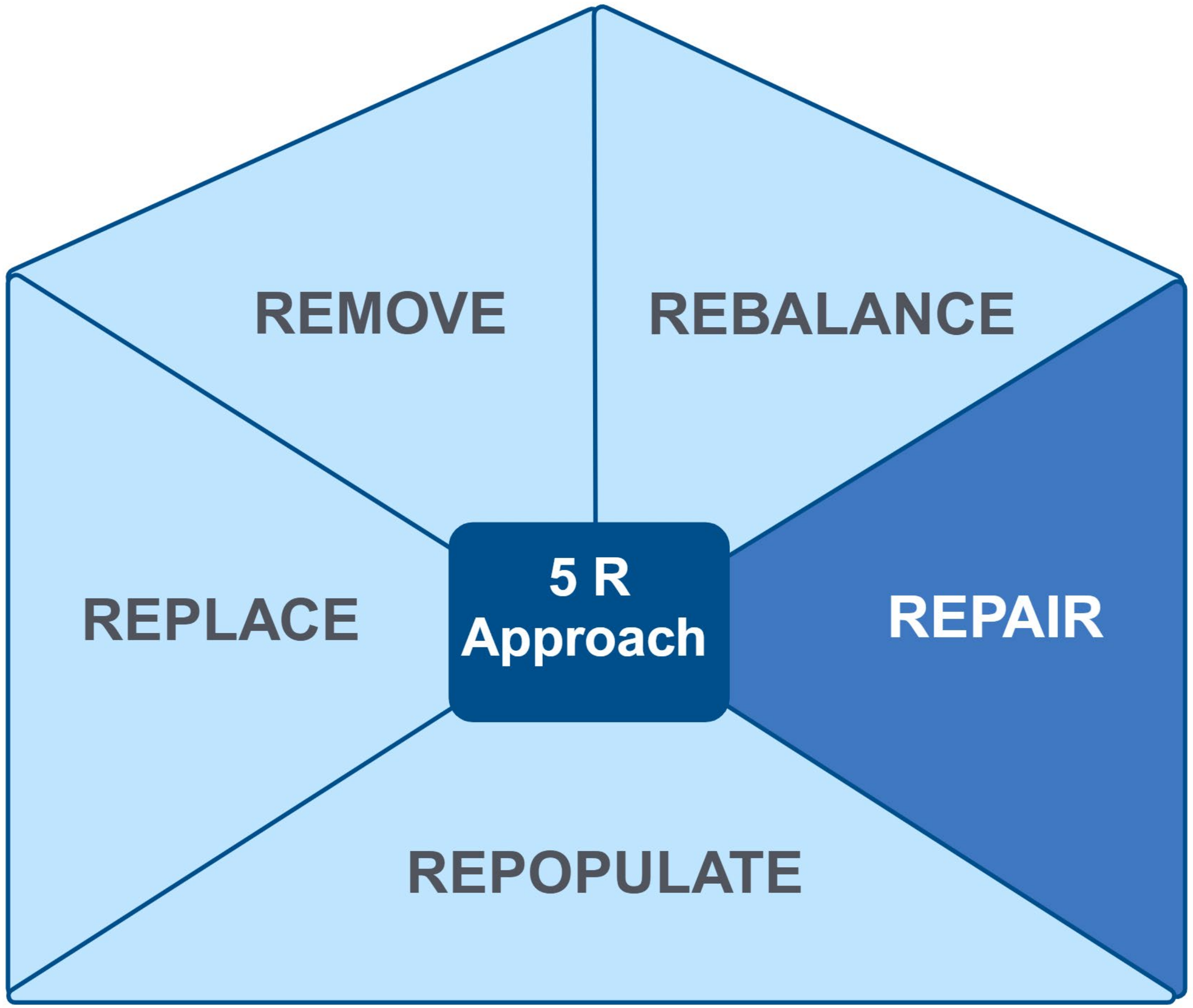
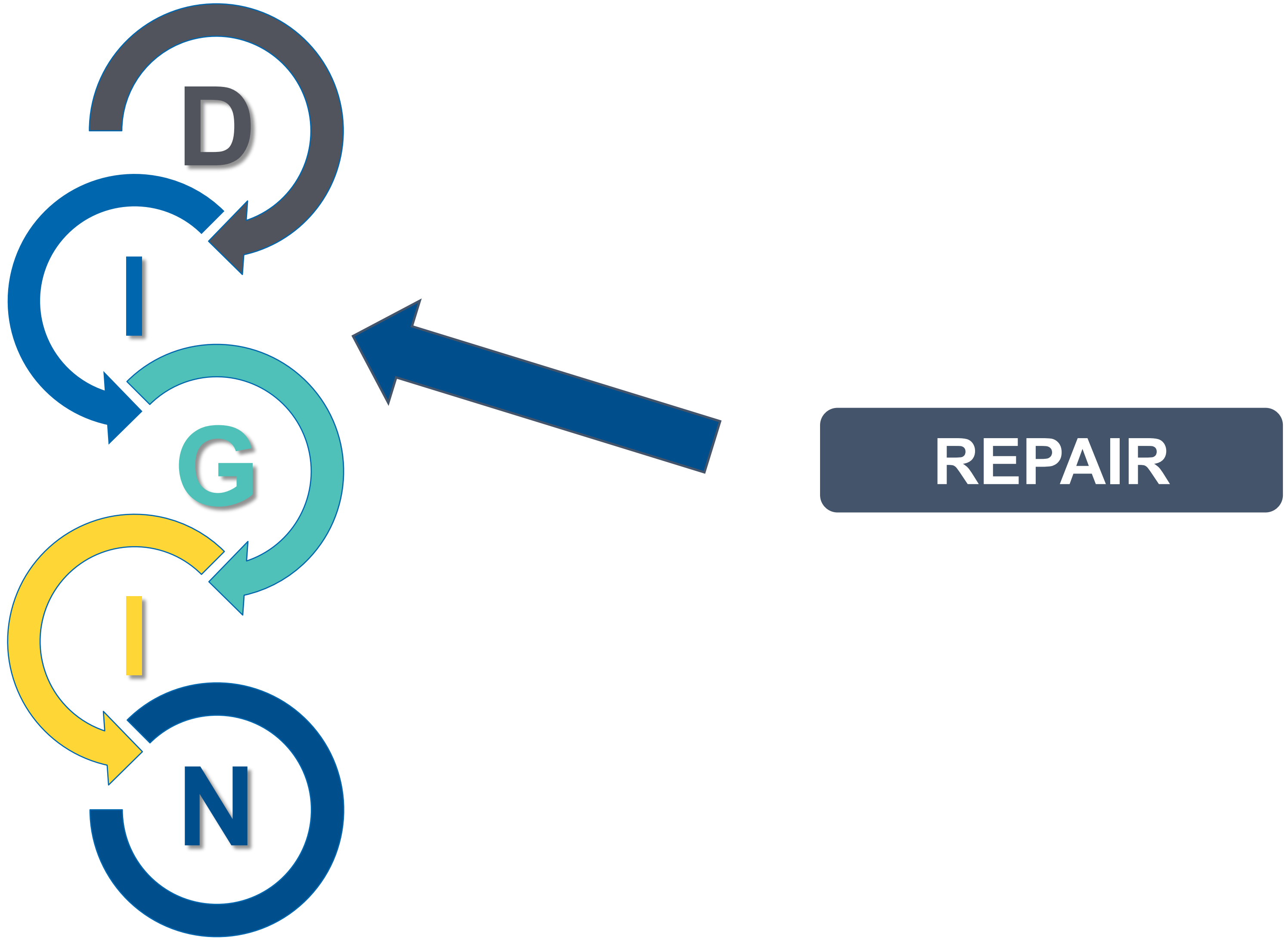


Image: Created with BioRender.com by The Institute for Functional Medicine

Long Description: The 5 R Approach

The 5 R Approach includes remove, replace, repopulate, repair, rebalance. Here the emphasis is on Repair.

Repair Interventions Are Most Relevant for Addressing Increased Intestinal Permeability and Brush Border Injury



The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Long Description: Repair Interventions

Repair interventions are most relevant or the I for D I G I N or intestinal permeability.

DIGIN Dysfunctions and Stool Test Biomarkers

Increased intestinal permeability cannot be reliably assessed with stool testing

Gut DIGIN Dysfunction

Maldigestion/ Malabsorption
- Pancreatic Elastase
- Valerate
- Iso-Valerate
- Iso-Butyrate
- Fecal Fat

Intestinal Permeability
- No reliable stool test biomarker

Gut Dysbiosis
- Bacteria
- Ova/Parasite
- Yeast/Fungi
- SCFA: Butyrate
- Valerate
- Propionate
- Acetate
- βGlucuronidase

Gut Inflammation
- Calprotectin
- Lactoferrin
- Eosinophil Protein X
- sIgA
- Lysozyme

ENS Dysfunction
- No reliable stool test biomarker

Long Description: D I G I N Dysfunctions and Stool Test Biomarkers

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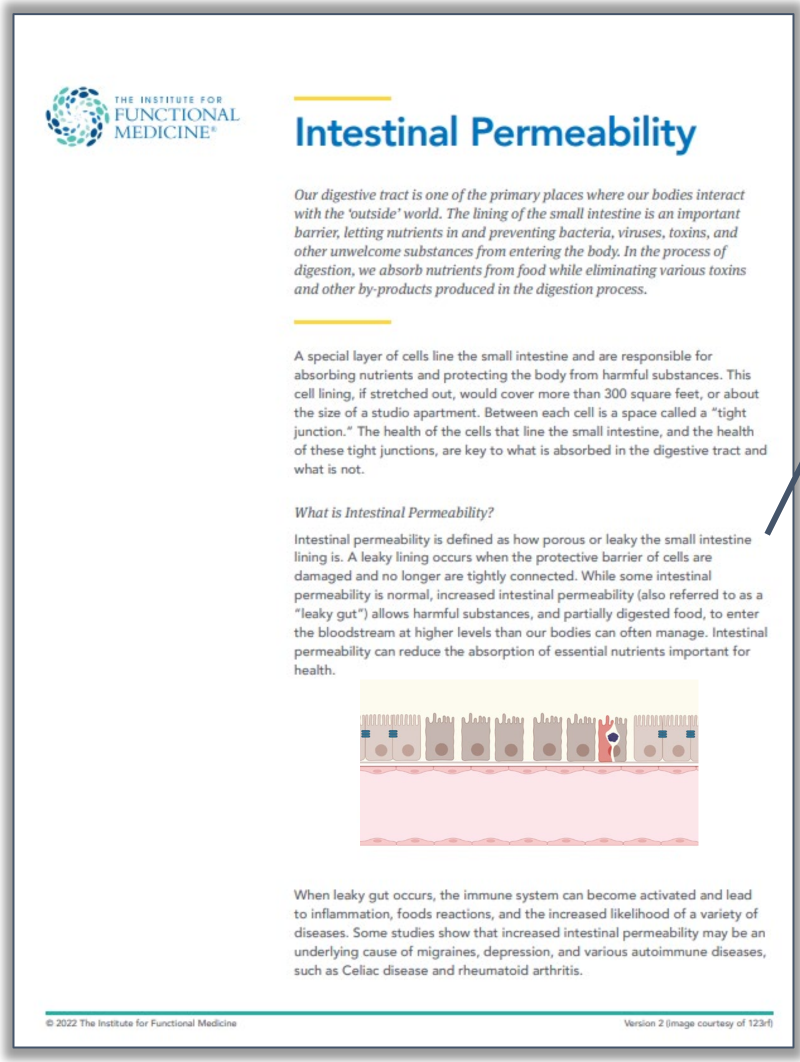
- Gut Inflammation: calprotectin, lactoferrin, Eosinophil Protein X, secretory I G A, lysozyme
- E N S Dysfunction: no reliable stool test biomarker
- Here we are emphasizing increased intestinal permeability cannot be reliably assessed with stool testing.

Repair the Gut Lining



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Intestinal Permeability

Our digestive tract is one of the primary places where our bodies interact with the 'outside' world. The lining of the small intestine is an important barrier, letting nutrients in and preventing bacteria, viruses, toxins, and other unwelcome substances from entering the body. In the process of digestion, we absorb nutrients from food while eliminating various toxins and other by-products produced in the digestion process.

A special layer of cells line the small intestine and are responsible for absorbing nutrients and protecting the body from harmful substances. This cell lining, if stretched out, would cover more than 300 square feet, or about the size of a studio apartment. Between each cell is a space called a "tight junction." The health of the cells that line the small intestine, and the health of these tight junctions, are key to what is absorbed in the digestive tract and what is not.

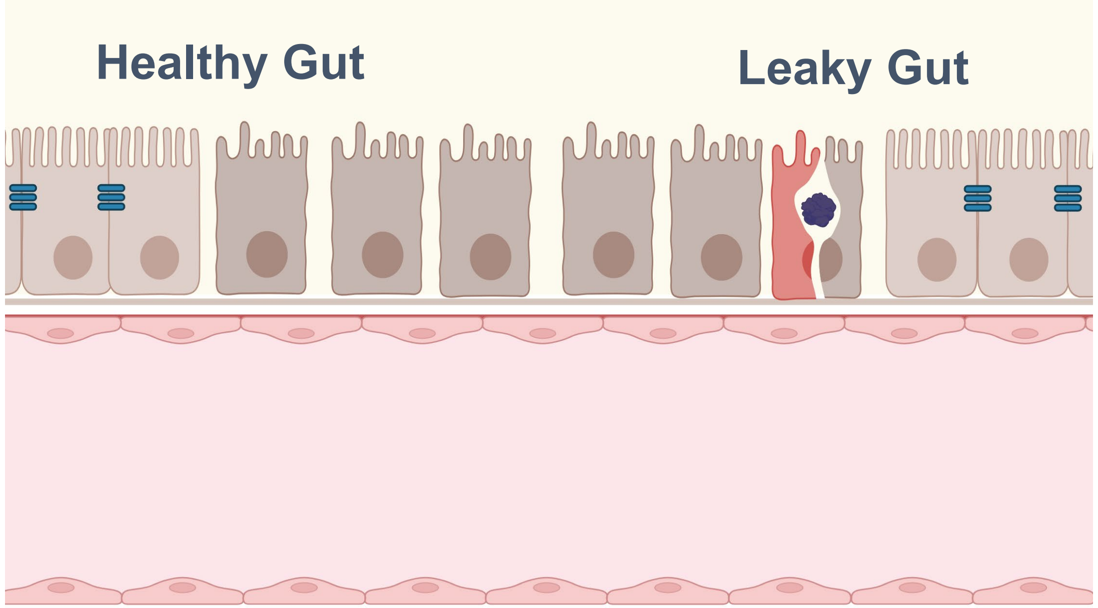
What is Intestinal Permeability?

Intestinal permeability is defined as how porous or leaky the small intestine lining is. A leaky lining occurs when the protective barrier of cells are damaged and no longer are tightly connected. While some intestinal permeability is normal, increased intestinal permeability (also referred to as a "leaky gut") allows harmful substances, and partially digested food, to enter the bloodstream at higher levels than our bodies can often manage. Intestinal permeability can reduce the absorption of essential nutrients important for health.

When leaky gut occurs, the immune system can become activated and lead to inflammation, foods reactions, and the increased likelihood of a variety of diseases. Some studies show that increased intestinal permeability may be an underlying cause of migraines, depression, and various autoimmune diseases, such as Celiac disease and rheumatoid arthritis.

Version 2 (image courtesy of 123rf)

Patient Educational Resource Available in your Toolkit



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Accessible version available in your toolkit.

Long Description: Repair the Gut Lining

- Toolkit Item: Intestinal Permeability

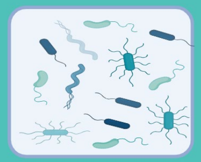
Practical Applications: Using the 5R Approach

- Does the patient need to have **Repair** of the GI lining?
- In lieu of stool testing, lines of evidence for systematic inflammation support these anti-inflammatory interventions:

Anti-Inflammatory Foundational Food/Lifestyle

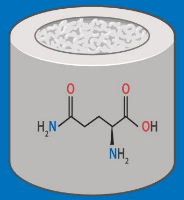

Core Food Plan


Phytonutrient Spectrum


Probiotics


Vitamin D


Omega-3s



L-Glutamine

Images: *Created with BioRender.com by The Institute for Functional Medicine*

Long Description: Practical Applications

Anti-Inflammatory foundational food and lifestyle approaches to repair the GI lining:

Core Food Plan

Phytonutrient Spectrum

Probiotics

Vitamin D

Omega-3's

L-Glutamine

Repair: Therapeutic Considerations

Foundational: Anti-Inflammatory Foods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Anti-Inflammatory Foods

Ongoing, low-grade inflammation underlies many common health problems. These include heart disease, type 2 diabetes, certain cancers, digestive issues, arthritis, and various brain disorders. Among the top causes of chronic inflammation are unhealthy eating, physical inactivity, and obesity (especially belly fat). Chronic infections, unbalanced gut bacteria, and toxin exposure also promote inflammation. A healthy lifestyle, including nutritious eating habits, helps tame inflammation.

How Foods Fight Inflammation

Whole foods like vegetables, fruits, whole grains, legumes, nuts, and seeds are rich in inflammation-fighting compounds. These include many vitamins, minerals, phytonutrients (plant compounds), and fiber. Omega-3 fats and probiotics (good bacteria), found in some animal and plant foods, can also calm inflammation.

The nutritional components of foods work in different ways to control inflammation. Some prevent the body from making as many inflammatory molecules. Others work as antioxidants to protect cells against damage from inflammation. Some promote a healthy community of gut bacteria, which can also help combat inflammation.

The following sections highlight foods with anti-inflammatory benefits, as shown in research studies. Similar foods in these categories could also have anti-inflammatory benefits but may be less well-studied.

Vegetables

Vegetables offer fiber and antioxidants, including beta-carotene, which gives an orange color. Vegetables are also super sources of green and red phytonutrients that fight inflammation. Some of the top anti-inflammatory vegetables are:

- **Dark green leafy vegetables:** lettuce (such as romaine and green leaf), spinach, Swiss chard
- **Cruciferous vegetables:** broccoli, broccoli sprouts, cabbage, cauliflower, collards, kale, mustard greens
- **Other vegetables:** beets, carrots, garlic, onions, tomatoes

INFLAMMATORY FOODS TO LIMIT OR AVOID

- Highly processed foods
- Fast food
- Products from grain-fed animals
- High omega-6 oils, such as corn and soybean
- Highly processed fats, like deep-fat fried
- Refined grains/carbs
- Refined sweeteners
- Soda
- Alcohol
- Any food you are sensitive to

Version 1

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Accessible version available in your toolkit.

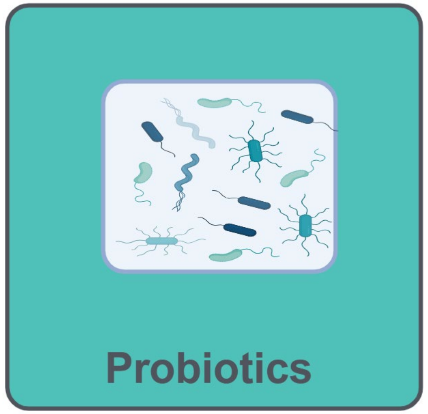
Long Description: Repair: Therapeutic Considerations

- Toolkit Items: Anti-Inflammatory foods, the Core Food Plan, and Phytonutrient Spectrum

Therapeutic Interventions for **Repair**

Probiotics

- Method of Action: Increased Intestinal Permeability
  - Regulate dysbiotic bacteria
  - Reduce gut-derived LPS production
  - Reduce inflammatory cytokines
  - Modulate intestinal mucosa



- Probiotics may significantly reduce intestinal permeability
  - Primarily combinations of two or more Bifidobacterium and Lactobacillus species

Foundational Therapeutics: Probiotic Rich Food Sources	Second-Tier: Supplemental Combination of Strains

- Moludi J, Maleki V, Jafari-Vayghyan H, Vaghef-Mehrabany E, Alizadeh M. Metabolic endotoxemia and cardiovascular disease: a systematic review about potential roles of prebiotics and probiotics. *Clin Exp Pharmacol Physiol*. 2020;47(6):927-939. doi:10.1111/1440-1681.13250
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- Image: Created with BioRender.com by The Institute for Functional Medicine

Therapeutic Interventions for **Repair**

Vitamin D

- **Vitamin D** decreases inflammation
  - Downregulates nuclear factor-kB (NF-kB) activity
  - Increases IL-10 production
  - Decreases IL-6, IL-12, IFN-c, and TNF-a production
  - Strengthens epithelial barrier and tight junctions

- **Sunlight exposure:** Guidelines suggest that most people can meet some of their vitamin D requirements via sunlight exposure, particularly between 10 am – 4 pm.
  - Dependent on: Latitude, season, age, cloud/smog coverage, skin melanin, and sunscreen use

- See: “References: Vitamin D Therapeutic Interventions”
- Image: Created with BioRender.com by The Institute for Functional Medicine

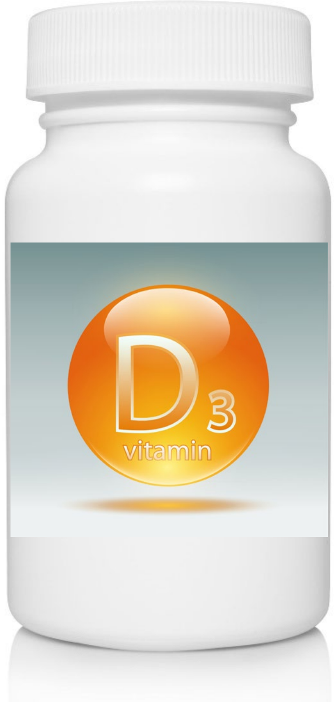
Repair: Vitamin D

Second-Tier Interventions: Supplemental

- **IBD relapse**: Vitamin D supplementation may reduce the risk of relapse by up to 52% in patients with IBD.
  - Deficiency in vitamin D may compromise the mucosal bar

Vitamin D: Supplement until optimal blood level of 25(OH)D3 is achieved

- Immune/inflammatory support: **50-80 ng/mL**



- See: "References: Vitamin D Therapeutic Interventions"

References

VITAMIN D THERAPEUTIC INTERVENTIONS

Vitamin D supplementation may reduce the risk of relapse in patients with IBD (Strong)

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- Wallace C, Gordon M, Sinopoulou V, Limketkai BN. Vitamin D for the treatment of inflammatory bowel disease. *Cochrane Database Syst Rev*. 2023;10(10):CD011806. doi:10.1002/14651858.CD011806.pub2

Vitamin D may reduce inflammation (CRP) (Strong)

- Guan Y, Hao Y, Guan Y, Bu H, Wang H. Effects of vitamin D supplementation on blood markers in ulcerative colitis patients: a systematic review and meta-analysis. Eur J Nutr. 2022;61(1):23-35. doi:10.1007/s00394-021-02603-2
- Ahamed Z R, Dutta U, Sharma V, et al. Oral Nano Vitamin D Supplementation Reduces Disease Activity in Ulcerative Colitis: A Double-Blind Randomized Parallel Group Placebo-controlled Trial. J Clin Gastroenterol. 2019;53(10):e409-e415. doi:10.1097/MCG.0000000000001233
- Guzman-Prado Y, Samson O, Segal JP, Limdi JK, Hayee B. Vitamin D Therapy in Adults With Inflammatory Bowel Disease: A Systematic Review and Meta-Analysis. Inflamm Bowel Dis. 2020;26(12):1819-1830. doi:10.1093/ibd/izaa087
- Jiang Q, Prabahar K, Saleh SAK, et al. The Effects of Vitamin D Supplementation on C-Reactive Protein and Systolic and Diastolic Blood Pressure in Postmenopausal Women: A Meta-Analysis and Systematic Review of Randomized Controlled Trials. J Acad Nutr Diet. 2024;124(3):387-396.e5. doi:10.1016/j.jand.2023.10.013]

Vitamin D Supplementation to Desired Lab Values

When possible, consider baseline testing and periodic monitoring of serum 25(OH)D to reach a targeted dose.

For optimal health in adults, desired lab value is
> 50 ng/mL to < 80 ng/mL

VITAMIN D, 25-HYDROXY, SERUM

F	25OH, VITAMIN D	49.9	32.0-100.0 (ng/mL)

```
- VITAMIN D,25-OH TEST INFORMATION
-
-                 Range (ng/mL)          Suggested Interpretation
-
-
-
-
```

VITAMIN D,25-OH,TOTAL,IA (17306)

F	VITAMIN D,25-OH,TOTAL,IA	75	30-100 (ng/mL)	CB

```
- Vitamin D Status          25-OH Vitamin D:
-
- Deficiency:                     <20 ng/mL
- Insufficiency:               20 - 29 ng/mL
- Optimal:                  > or = 30 ng/mL
-
- For 25-OH Vitamin D testing on patients on
- D2-supplementation and patients for whom quantitation
- of D2 and D3 fractions is required, the QuestAssureD(TM)
- 25-OH VIT D, (D2,D3), LC/MS/MS is recommended: order
```


See: "References: Vitamin D Laboratory Levels"

Lab images courtesy of Elizabeth McMillan, CNS, LDN

Long Description: Vitamin D Supplementation

This image depicts and increase in baseline 25 O H Vitamin D level from 49.9 N G per M L to 75 N G per M L.

References

VITAMIN D LABORATORY LEVELS

- Anitua E, Tierno R, Alkhraisat MH. Current opinion on the role of vitamin D supplementation in respiratory infections and asthma/COPD exacerbations: a need to establish publication guidelines for overcoming the unpublished data. *Clin Nutr*. 2022;41(3):755-777. doi:10.1016/j.clnu.2022.01.029
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Therapeutic Interventions for **Repair**

Omega-3 Fatty Acids (FA)

- Omega-3 FA, specifically EPA (eicosatetraenoic acid) and DHA (docosahexaenoic acid) repair inflammation and increased intestinal permeability via:
  - Reducing pro-inflammatory cytokines
  - Strengthening tight junctions
  - Enhancing mucosal integrity
  - Modulating gut microbiota
  - Activating specialized pro-resolving lipid mediators (SPMs)

Foundational: Food Sources

Secondary Therapeutic Dosage: 2-4 grams of combined EPA and DHA

- See: “References: Omega-3 Fatty Acids for Anti-Inflammatory Foundations”
- Created with BioRender.com by The Institute for Functional Medicine

References

OMEGA-3 FATTY ACIDS FOR ANTI-INFLAMMATORY FOUNDATIONS

Omega-3 supplementation reduces CRP (Strong)

- Kavyani Z, Musazadeh V, Fathi S, Hossein Faghfouri A, Dehghan P, Sarmadi B. Efficacy of the omega-3 fatty acids supplementation on inflammatory biomarkers: An umbrella meta-analysis. Int Immunopharmacol. 2022;111:109104. doi:10.1016/j.intimp.2022.109104
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Additional Omega-3 Citations

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In Your Toolkit

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Food Sources:

Essential Fatty Acids

Essential Fatty Acids (EFAs) cannot be created by your body. They are called "essential" because you must get them through your diet. There are two primary families of essential fatty acids: omega-6 and omega-3.

Linoleic acid (LA) and arachidonic acid (AA) are two common omega-6 fatty acids. LA is found in most plant oils, and AA is present in meat, poultry, and eggs. Alpha-linolenic acid (ALA), eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA) are the most common omega-3 fatty acids. Some plant foods contain ALA, while fatty fish are the best source of EPA and DHA.

Every cell in the body needs EFAs for proper structure and function. EFAs promote healthy nerve functioning and play a role in inflammation.

The Adequate Intakes (AIs) for omega-6 fatty acids are as follows:

- **Females, ages 19-50:** 12 grams per day
- **Females, ages 51+:** 11 grams per day
- **Pregnancy, lactating:** 13 grams per day
- **Males, ages 19-50:** 17 grams per day
- **Males, ages 51+:** 14 grams per day

Food, Standard Serving Size	Amount of LA Omega-6s (in grams)
Walnuts, 1 ounce (14 halves)	10.8
Safflower oil, 1 Tbsp	10.1
Sunflower seeds (oil roasted), 1 ounce (¼ cup)	9.7
Grapeseed oil, 1 Tbsp	9.5
Pine nuts, 1 ounce (¼ cup)	9.4
Sunflower oil, 1 Tbsp	8.9
Hemp seeds (hulled), 3 Tbsp	8.3
Pecans (oil roasted), 1 ounce (19 halves)	6.4
Pumpkin seeds (dried), 1 ounce (¼ cup)	5.9
Sesame oil, 1 Tbsp	5.6

The Adequate Intakes (AIs) for omega-3 fatty acids are as follows:

- **Females, ages 19+:** 1.1 grams per day
- **Pregnancy:** 1.4 grams per day
- **Lactating:** 1.3 grams per day
- **Males, ages 19+:** 1.6 grams per day

Food, Standard Serving Size	Amount of Omega-3s (in grams)
Flaxseed oil, 1 Tbsp	7.3 (ALA)
Chia seeds, 1 ounce (2 Tbsp)	5.1 (ALA)
Hemp seeds (hulled), 3 Tbsp	2.6 (ALA)
Walnuts, 1 ounce (14 halves)	2.6 (ALA)
Flaxseed (whole), 1 Tbsp	2.4 (ALA)
Atlantic salmon (cooked), 3 ounces	1.8 (EPA and DHA)
Atlantic herring (cooked), 3 ounces	1.7 (EPA and DHA)
Sardines (canned), 3 ounces	1.2 (EPA and DHA)
Atlantic mackerel (cooked), 3 ounces	1.0 (EPA and DHA)

REFERENCES

1. U.S. Department of Health and Human Services, National Institutes of Health, Office of Dietary Supplements. Omega-3 fatty acids. https://ods.od.nih.gov/factsheets/Omega3FattyAcids-HealthProfessional/. Accessed April 2, 2024.
2. U.S. Department of Agriculture, Agricultural Research Service, Nutrient Data Laboratory. USDA National Nutrient Database for Standard Reference, Legacy Version. https://fdc.nal.usda.gov/. Accessed April 2, 2024.
3. Oregon State University, Linus Pauling Institute, Micronutrient Information Center. Essential fatty acids. https://lpi.oregonstate.edu/mic/other-nutrients/essential-fatty-acids#introduction. Reviewed May 2014. Accessed April 2, 2024.

Version 3; updated 2024

Food, Standard Serving Size	Amount of Omega-3s (in grams)
Flaxseed oil, 1 Tbsp	7.3 (ALA)
Chia seeds, 1 ounce (2 Tbsp)	5.1 (ALA)
Hemp seeds (hulled), 3 Tbsp	2.6 (ALA)
Walnuts, 1 ounce (14 halves)	2.6 (ALA)
Flaxseed (whole), 1 Tbsp	2.4 (ALA)
Atlantic salmon (cooked), 3 ounces	1.8 (EPA and DHA)
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Sardines (canned), 3 ounces	1.2 (EPA and DHA)
Atlantic mackerel (cooked), 3 ounces	1.0 (EPA and DHA)

Accessible version of this resource in your toolkit.

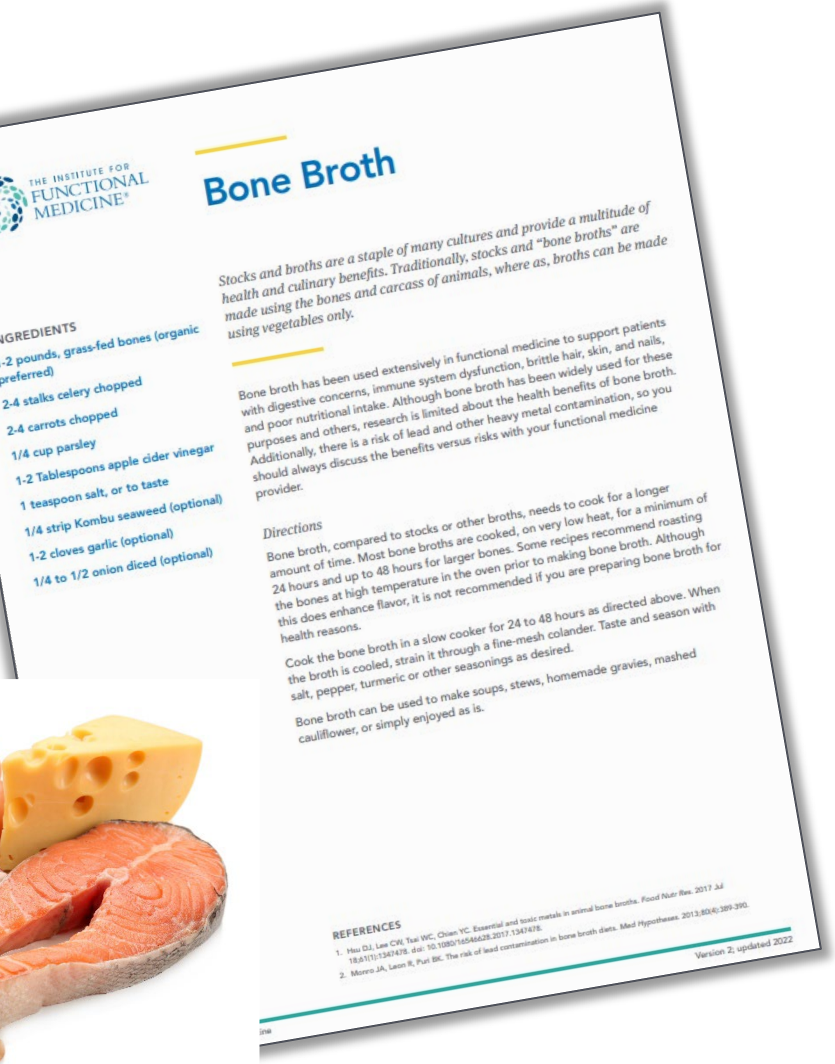
Long Description: In Your Toolkit

- Toolkit Item: Essential Fatty Acids

<u>Repair</u>: L-Glutamine Therapeutic Considerations

Foundational: L-Glutamine-Rich Foods

- Eggs
- Leafy vegetables
- Dairy products
- Organ meat
- Chicken
- Asparagus
- Parsley
- Seaweed
- Beef
- Nuts
- Legumes
- Tofu
- Rice
- Beans
- Cabbage
- Grains



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Bone Broth

Stocks and broths are a staple of many cultures and provide a multitude of health and culinary benefits. Traditionally, stocks and "bone broths" are made using the bones and carcass of animals, where as, broths can be made using vegetables only.

INGREDIENTS

- 1-2 pounds, grass-fed bones (organic preferred)
- 2-4 stalks celery chopped
- 2-4 carrots chopped
- 1/4 cup parsley
- 1-2 Tablespoons apple cider vinegar
- 1 teaspoon salt, or to taste
- 1/4 strip Kombu seaweed (optional)
- 1-2 cloves garlic (optional)
- 1/4 to 1/2 onion diced (optional)

Bone broth has been used extensively in functional medicine to support patients with digestive concerns, immune system dysfunction, brittle hair, skin, and nails, and poor nutritional intake. Although bone broth has been widely used for these purposes and others, research is limited about the health benefits of bone broth. Additionally, there is a risk of lead and other heavy metal contamination, so you should always discuss the benefits versus risks with your functional medicine provider.

Directions

Bone broth, compared to stocks or other broths, needs to cook for a longer amount of time. Most bone broths are cooked, on very low heat, for a minimum of 24 hours and up to 48 hours for larger bones. Some recipes recommend roasting the bones at high temperature in the oven prior to making bone broth. Although this does enhance flavor, it is not recommended if you are preparing bone broth for health reasons.

Cook the bone broth in a slow cooker for 24 to 48 hours as directed above. When the broth is cooled, strain it through a fine-mesh colander. Taste and season with salt, pepper, turmeric or other seasonings as desired.

Bone broth can be used to make soups, stews, homemade gravies, mashed cauliflower, or simply enjoyed as is.

REFERENCES

1. Hsu DJ, Lee CW, Tsai WC, Chien YC. Essential and toxic metals in animal bone broths. Food Nutr Res. 2017 Jul 18;61(1):1347478. doi: 10.1080/16546628.2017.1347478.
2. Monro JA, Leon R, Puri BK. The risk of lead contamination in bone broth diets. Med Hypotheses. 2013;80(4):389-390.

Version 2; updated 2022

Accessible version of this resource in your toolkit.

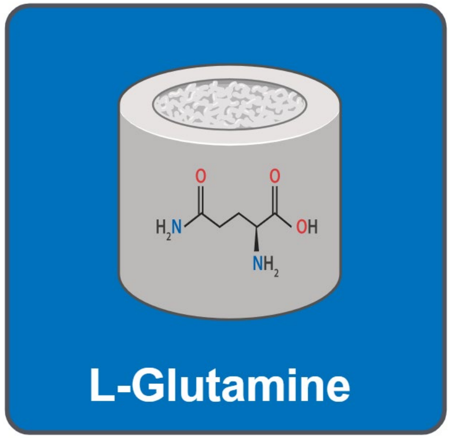
Long Description: **Repair**: L-Glutamine Therapeutic Considerations

- Toolkit Item: Bone Broth

Therapeutic Interventions for **Repair**

L-Glutamine

- Functions for Anti-Inflammatory Support
  - Preferred fuel for enterocytes of small intestine
  - Increases intestinal villous height
  - Stimulates gut mucosal cellular proliferation
  - Maintains mucosal integrity
  - Prevents intestinal permeability and bacterial translocation



Second-Tier

Therapeutic Dosage: 5g by mouth 3x/day for intestinal support

See: "References: L-Glutamine for Anti-Inflammatory Foundations"

References

L-GLUTAMINE FOR ANTI-INFLAMMATORY FOUNDATIONS

L-glutamine may normalize intestinal hyperpermeability in patients with IBS-D (Moderate)

- Zhou Q, Verne ML, Fields JZ, et al. Randomised placebo-controlled trial of dietary glutamine supplements for postinfectious irritable bowel syndrome. *Gut*. 2019;68(6):996-1002. doi:10.1136/gutjnl-2017-315136

Additional L-glutamine citations

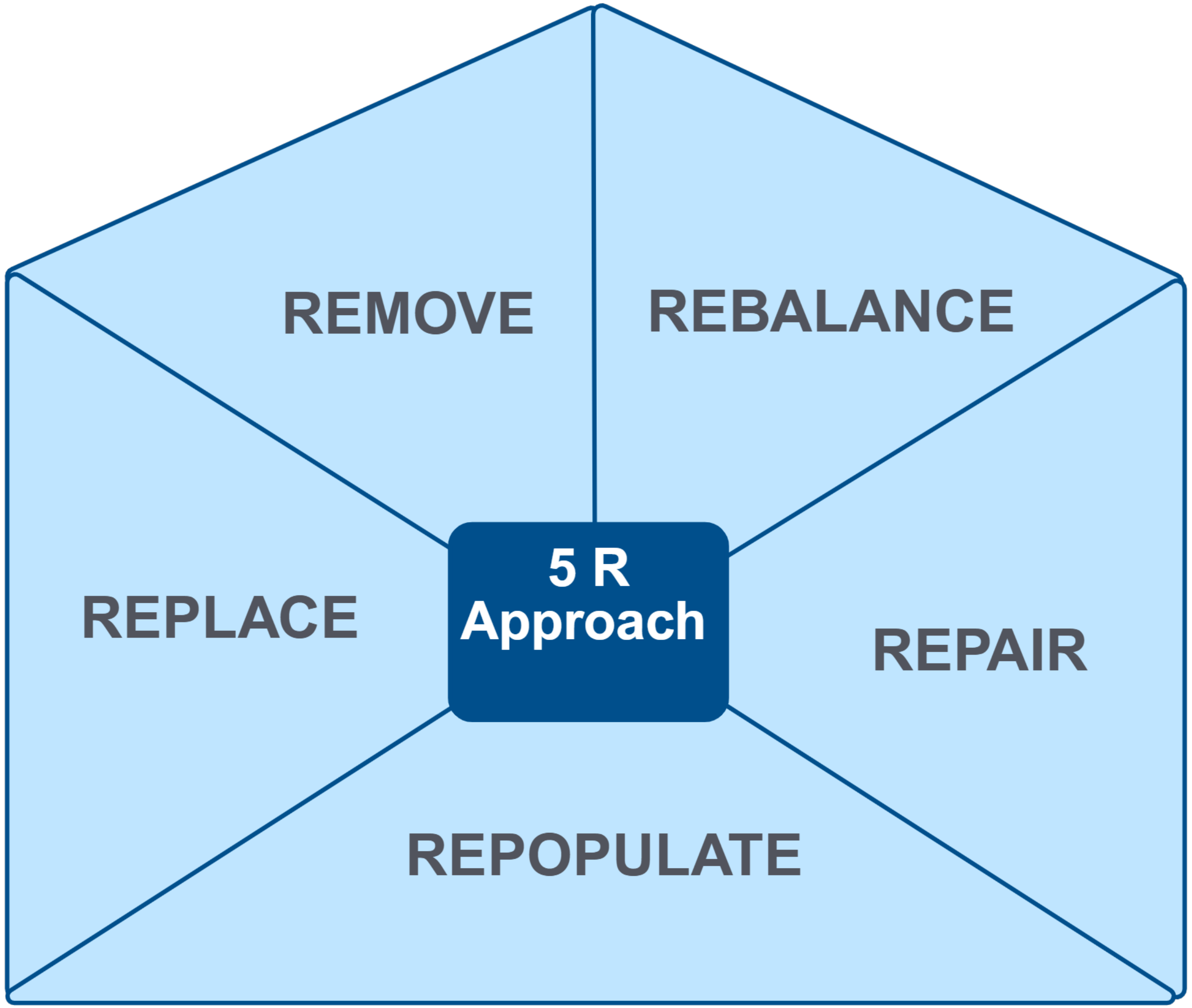
- Wang B, Pei J, Xu S, Liu J, Yu J. A glutamine tug-of-war between cancer and immune cells: recent advances in unraveling the ongoing battle [published correction appears in *J Exp Clin Cancer Res*. 2024;43(1):93]. *J Exp Clin Cancer Res*. 2024;43(1):74. doi:10.1186/s13046-024-02994-0
- Kjaer M, Frederiksen AKS, Nissen NI, et al. Multinutrient supplementation increases collagen synthesis during early wound repair in a randomized controlled trial in patients with inguinal hernia. *J Nutr*. 2020;150(4):792-799. doi:10.1093/jn/nxz324

Foundational Therapeutic Approach: 5R Therapeutics

LIFESTYLE FIRST

FOOD FIRST

GUT FIRST



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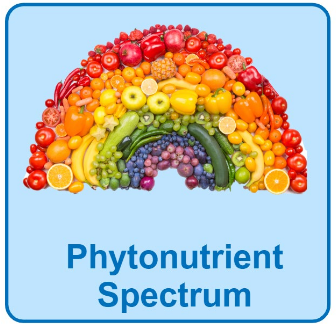
Long Description: The 5 R Approach

The 5 R Approach includes remove, replace, repopulate, repair, rebalance.

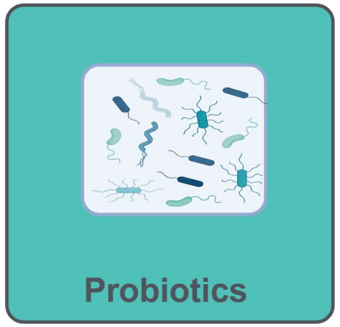
Using the 5R Approach

Anti-Inflammatory Foundation Summary

Dietary



Second-Tier Supplemental Support

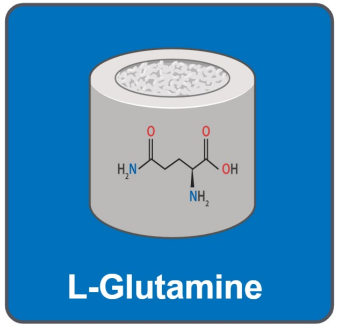


Probiotics: Primarily combinations of two or more Bifidobacterium and Lactobacillus species

Omega-3: 2-4 grams of combined EPA and DHA

Vitamin D: Supplement until optimal blood level of 25(OH)D3 is achieved

- Immune/inflammatory support: **50-80 ng/mL**

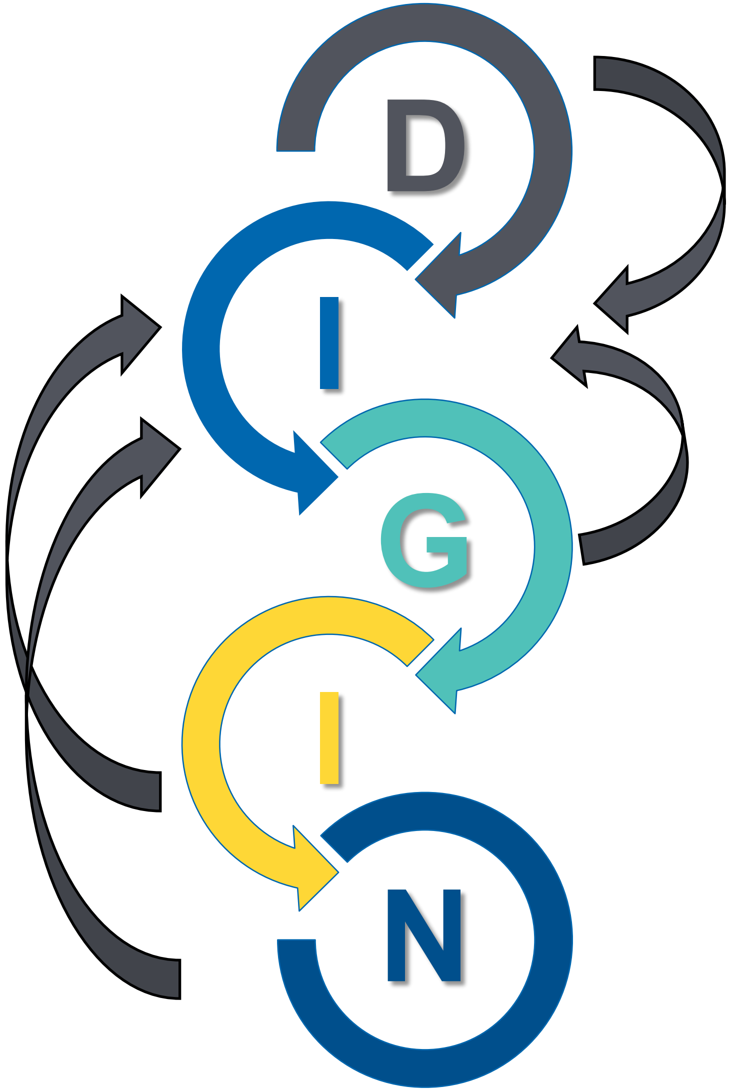


L-glutamine: 5g 3x/day for intestinal support

Long Description: Using the 5 R Approach

Anti-Inflammatory Foundation Summary
Dietary: Core Food Plan and Phytonutrient Spectrum
Second-Tier Supplemental Support: Probiotics, Vitamin D, Omega-3s, L-Glutamine

All Other Gut Dysfunctions of the DIGIN Model May be Mediators of Increased Intestinal Permeability



Therefore, whenever increased intestinal permeability is suspected or identified, it is imperative to evaluate the patient for the other DIGIN dysfunctions.

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Mediators of Increased Intestinal Permeability

Rationale for Considering All 5 of the 5Rs

- **R**emove ← Meds, toxins, problem foods damage intestine
- **R**eplace ← Deficiencies of nutrients that are essential for integrity of tight junctions
- **R**epopulate ← Dysbiotic microbes are associated with increased intestinal permeability
- **R**epair ← Nutrients, nutraceuticals, botanicals
- **R**ebalance ← Stress, inadequate sleep, overexercise, all mediate dysregulation of tight junctions

Long Description: Mediators of Increased Intestinal Permeability

Mediators of Maldigestion and Malabsorption for all five of the five R's.

Remove: medications, toxins, problem foods that are damaging the intestine

Replace: Deficiencies of nutrients that are essential for integrity of tight junctions

Repopulate: Dysbiotic microbes are associated with increased intestinal permeability

Repair: Nutrients, nutraceuticals, botanicals

Rebalance: Stress, inadequate sleep, overexercise, all mediate dysregulation of tight junctions

Case Vignette

- 32 yo female with pre-diabetes, obesity, 9 months postpartum develops severe right upper quadrant pain radiating to her back after she eats pepperoni pizza
- An ultrasound shows gallstones and a HIDA scan confirms biliary dyskinesia
- The patient wants to avoid surgery if possible

Is it most likely…?

- Hypochlorhydria
- Pancreatic Exocrine Dysfunction
- Bile Acid Disorder
- Brush Border Injury
- Colon Dysbiosis
- Gut Inflammation
- Increased Intestinal Permeability

Clinical Focus

- Summary and Conclusion – Key Takeaways

Resource Available in Your Toolkit

THE INSTITUTE FOR FUNCTIONAL MEDICINE

The 5R Framework for Gut Restoration

A properly functioning digestive system is critical to good health. Problems with the gastrointestinal (GI) tract can cause more than just stomach upset, gas, bloating, and diarrhea. GI issues may underlie chronic health problems that seem unrelated to digestive health, including autoimmune diseases (such as rheumatoid arthritis), skin problems (such as eczema and acne rosacea), and heart disease, among others.

In the bigger picture, how can we deal with all that can go wrong in the gut? In functional medicine, we use a program that goes by the simple acronym of the 5Rs: remove, replace, reinoculate, repair, and rebalance. When applied to various chronic health issues, the 5R program can lead to a dramatic improvement in symptoms and sometimes even complete resolution. The elements of the 5R program are described briefly here.

1. Remove

Remove stressors. Get rid of things that negatively affect the environment of the GI tract, including foods you are sensitive to, parasites, and potential problematic bacteria or yeast. This might involve using an elimination diet to find out what foods are triggering GI symptoms. It may also involve taking medications or herbs to get rid of an infection.

2. Replace

Replace digestive secretions. Add back things like digestive enzymes, bile acids, and hydrochloric acid that are required for proper digestion and that may be compromised by diet, medications, diseases, aging, or other factors.

3. Reinoculate

Help beneficial bacteria flourish by eating probiotic-rich foods or taking supplements that contain "good" bacteria, such as *Bifidobacteria* and *Lactobacillus* species, and by eating high-fiber foods that supply prebiotics to nourish good microbes.

- Probiotics are beneficial microorganisms found in the gut and are also called "friendly bacteria." The use of antibiotics kills both good and bad bacteria. Probiotics from supplements or food are often needed to help

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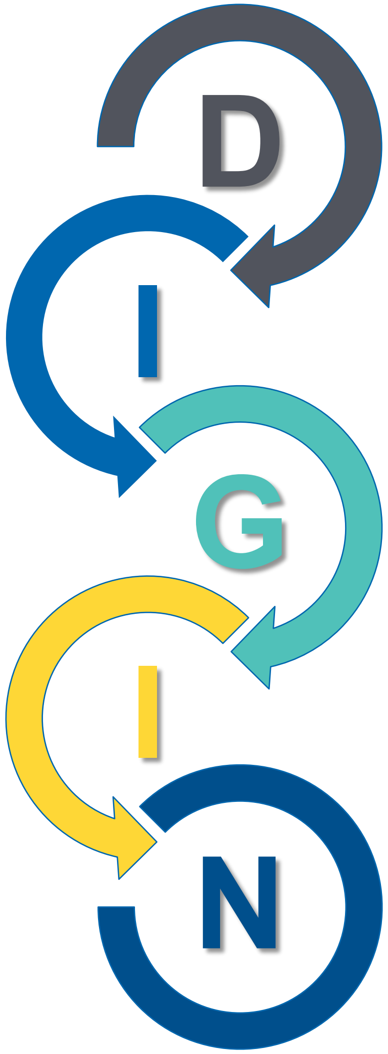
Long Description: Resource Available in your Toolkit

- Toolkit Item: The 5 R Framework for Gut Restoration

Summary: Main Correlations Between Each of the DIGIN Dysfunctions and 5R Interventions

- **R**eplace
- **R**epair
- **R**epopulate
- **R**emove
- **R**ebalance

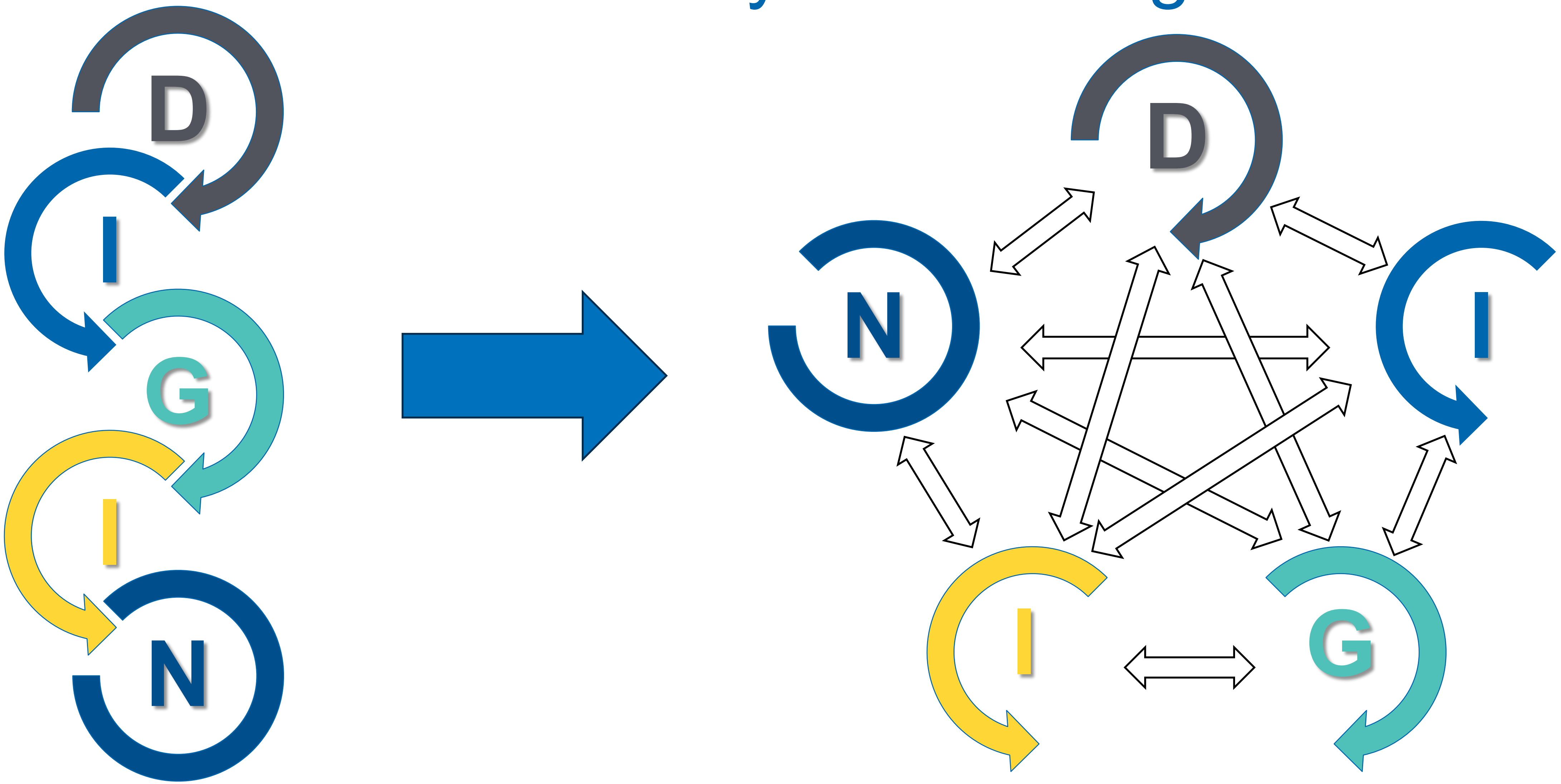


The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Long Description: Main Correlations Between Each of the D I G I N Dysfunctions and 5 R Interventions

The correlations between each D I G I N Dysfunctions and the 5 R Interventions – Replace, Repair, Repopulate, Remove, and Rebalance

DIGIN Dysfunctions: Mutually Reinforcing



The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Long Description: D I G I N Dysfunctions: Mutually Reinforcing

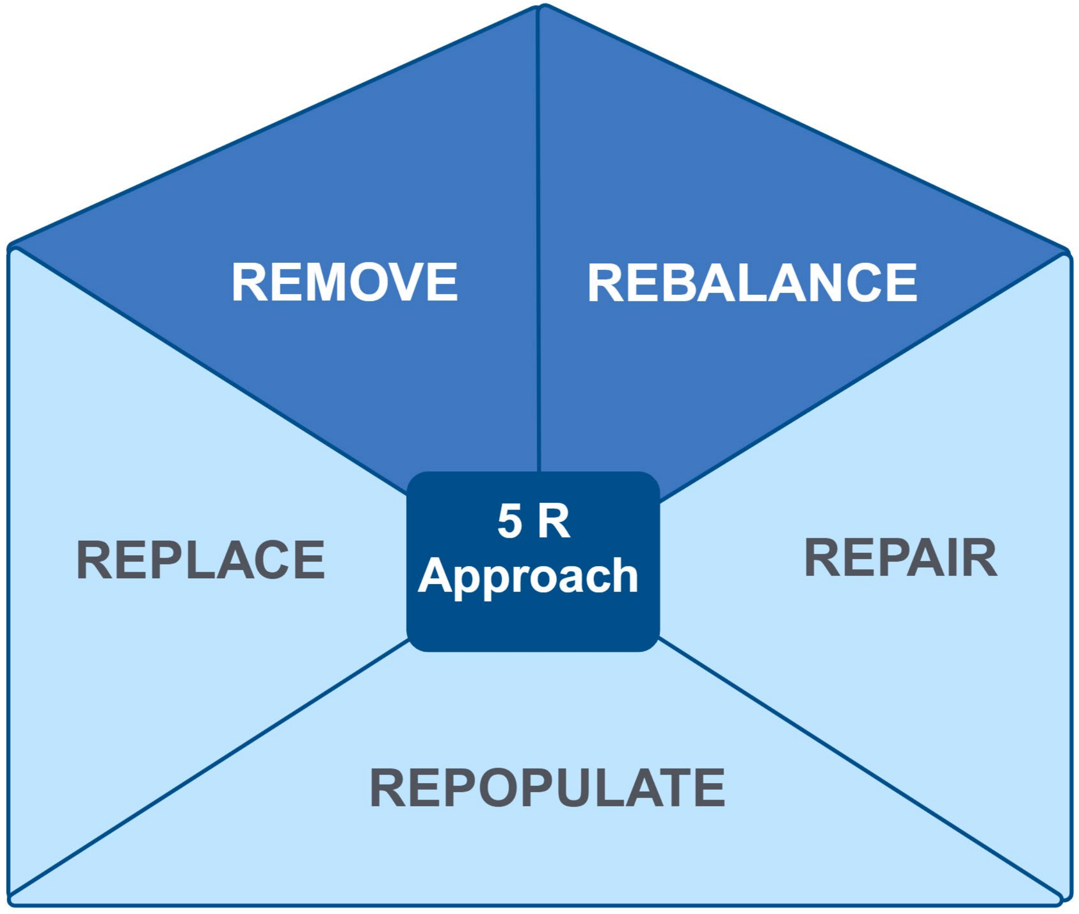
- Each D I G I N Dysfunction interconnects with the other.

Usual Sequence of the 5R Program

Primacy of the Remove and Rebalance Interventions

- **R**emove
- **R**eplace
- **R**epopulate
- **R**epair
- **R**ebalance

Remove & Rebalance interventions may be empirically initiated while waiting for the results of stool testing and other evaluative and/or therapeutic interventions.



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Long Description: Usual Sequence of the 5 R Program

The 5 R Approach includes remove, replace, repopulate, repair, rebalance. Here the emphasis is on Remove and Rebalance. Remove and Rebalance interventions may be empirically initiated while waiting for the results of stool testing and other evaluative and/or therapeutic interventions.

Foundational **Remove** Intervention

IFM's Core Food Plan

- Inflammatory elements
- Inconsistent behaviors
- Imbalanced fats
- High-glycemic impact
- Ultra-processed

Nutrient-Depleted Diet

Core Food Plan

- Nutrient-dense
- Healthy, whole foods
- Phytonutrient rainbow
- Balanced dietary choices

Long Description: Foundational **Remove** Intervention

The Core Food Plan emphasizes nutrient-dense, healthy whole foods, the phytonutrient rainbow, and balanced dietary choices. By following the Core Food Plan, patients will remove a nutrient-depleted diet that includes inflammatory elements, inconsistent behaviors, imbalanced fats, high-glycemic impact, and ultra-processed foods.

Rebalance: Foundational Interventions
(Must Be Personalized – Refer as Necessary)

Rebalance interventions support restorative processes in a patient's physiology.

Examples of Common Lifestyle, Mental, Emotional, and Spiritual Mediators of Gut Dysfunction

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships
• Adverse sleep behaviors	• Over exercise • Sedentarism	• Adverse meal behaviors	• Avoidable sources of stress	• Isolation, loneliness

See: "References: Rebalance"

Long Description: **Rebalance**: Foundational Interventions

- Common examples of modifiable lifestyle factors.
- Sleep: adverse sleep behaviors
- Physical Activity: over exercise; sedentarism
- Nutrition: adverse meal behaviors
- Stressors: avoidable sources of stress
- Relationships: isolation, loneliness

Foundational Interventions to Support Gastric Acidity

Best With a Protein-Containing Meal:

- Bitters
  - Examples: Ginger, wormwood, cinnamon, myrrh, orange peel, fennel, dandelion root, artichoke, rhubarb
- Vinegar
- Umeboshi plums
- Digestive enzymes with acid pH range

Lifestyle:

- Stress management
- Acupuncture

See: “References: Supporting Gastric Acidity”

Food-Derived Enzymes

- **Bromelain** (from pineapple)
  - Enzymatic breakdown of protein and is associated with reduced inflammation
- **Papain** (from papaya) – *Second Tier Intervention*
  - Dose: typically chewable: 100-250 mg

See: "References: Food-Derived Enzymes"

Current FDA-Approved “PERTs”

Pancreatic Enzyme Replacement Therapies

Second-Tier Therapeutics

Brand	Units of lipase (USP)
Creon	3,000; 6,000; 12,000; 24,000; 36,000
Zenpep	3,000; 5,000; 10,000; 15,000; 20,000; 25,000
Pancreaze	4,200; 10,500; 16,800; 21,000
Ultresa	13,800; 20,700; 23,000
Viokase	10,440; 20,880 (requires acid suppression)
Pertzye	8,000; 16,000

- Adapted from: Struyvenberg MR, Martin CR, Freedman SD. Practical guide to exocrine pancreatic insufficiency – breaking the myths. *BMC Med*. 2017;15(1):29. doi:10.1186/s12916-017-0783-y
- Licensed under CC BY 4.0.
- Perbtani Y, Forsmark CE. Update on the diagnosis and management of exocrine pancreatic insufficiency. *F1000Res*. 2019;8:F1000 Faculty Rev-1991. doi:10.12688/f1000research.20779.1

Pancreatin: Animal-Derived

TAKE WITH FOOD

Second-Tier therapeutics

Pancreatin (serving size 3 tablets)

Protease 156,000 USP
Amylase 156,000 USP
Lipase 24,960 USP

Pancreatin 8X 125 mg (1 tablet)

Amylase 25,000 USP units
Protease 25,000 USP units
Lipase 2,000 USP units

Pancreatin Range: 500-2,500 units/kg/meal

Pancreatin 4X 500 mg (1 tablet)

Amylase 50,000 USP units
Protease 50,000 USP units
Lipase 4,000 USP units

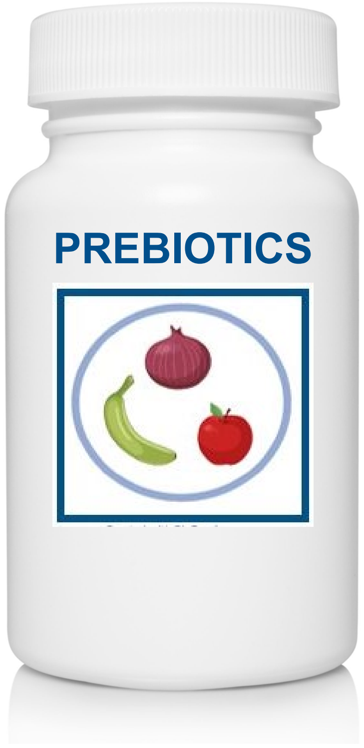
See: "References: Pancreatin: Animal-Derived"

Feeding the Microbiome: Prebiotics

Second-Tier Intervention: Supplemental Prebiotics

Treatment time – indefinite

- **FOS/fructo-oligosaccharides:** 1,000-5,000 mg, QD-TID



- See: "References: Prebiotics"
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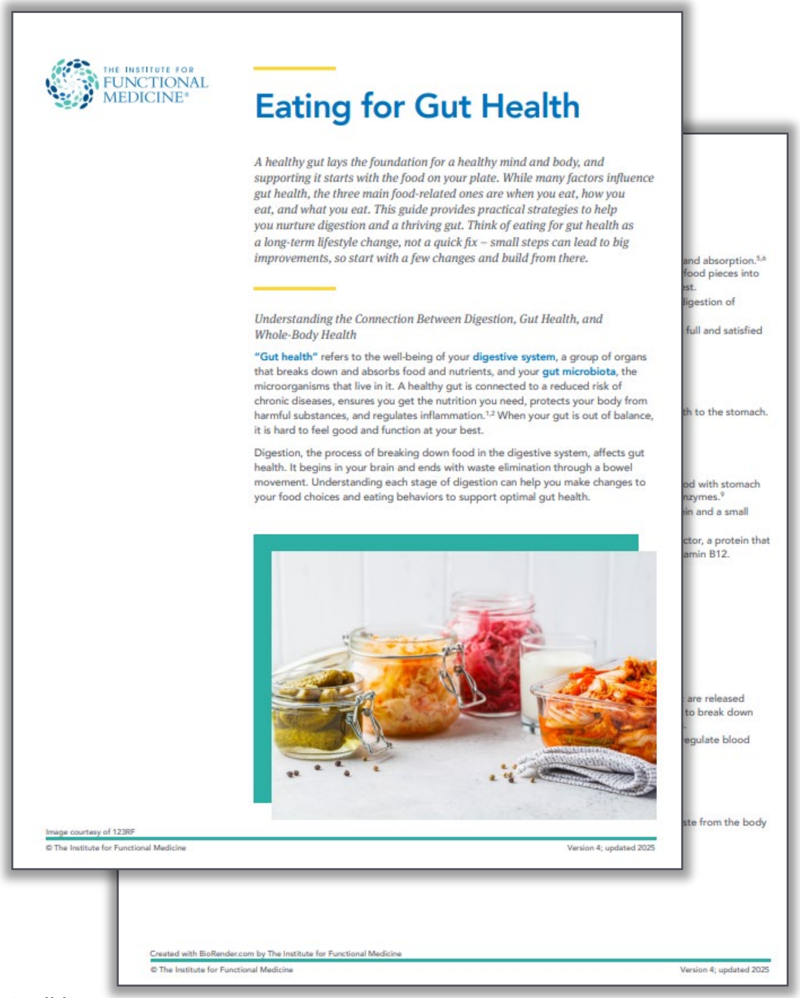
Feeding the Microbiome: Dietary Fiber Intake

- Paleolithic (estimated) intake **100-125 g/day**
- Recommended daily intake **25-35 g/day**
- Actual (average) intake in US **8-15 g/day**
- **Physiologic Properties**
  1. Slows transit in small bowel
  2. Increases stool bulk
  3. Holds on to water
  4. Increases estrogen excretion
  5. Binds minerals and organotoxins
  6. Stimulates bacterial growth
  7. Metabolized to SCFA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Eating for Gut Health

A healthy gut lays the foundation for a healthy mind and body, and supporting it starts with the food on your plate. While many factors influence gut health, the three main food-related ones are when you eat, how you eat, and what you eat. This guide provides practical strategies to help you nurture digestion and a thriving gut. Think of eating for gut health as a long-term lifestyle change, not a quick fix – small steps can lead to big improvements, so start with a few changes and build from there.

Understanding the Connection Between Digestion, Gut Health, and Whole-Body Health

"Gut health" refers to the well-being of your **digestive system**, a group of organs that breaks down and absorbs food and nutrients, and your **gut microbiota**, the microorganisms that live in it. A healthy gut is connected to a reduced risk of chronic diseases, ensures you get the nutrition you need, protects your body from harmful substances, and regulates inflammation.[1,2] When your gut is out of balance, it is hard to feel good and function at your best.

Digestion, the process of breaking down food in the digestive system, affects gut health. It begins in your brain and ends with waste elimination through a bowel movement. Understanding each stage of digestion can help you make changes to your food choices and eating behaviors to support optimal gut health.

Image courtesy of 123RF

Version 4; updated 2025

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Version 4; updated 2025

Accessible version available in your toolkit.

Long Description: **Feeding** the Microbiome: Dietary Fiber Intake

- Toolkit Item: Eating for Gut Health

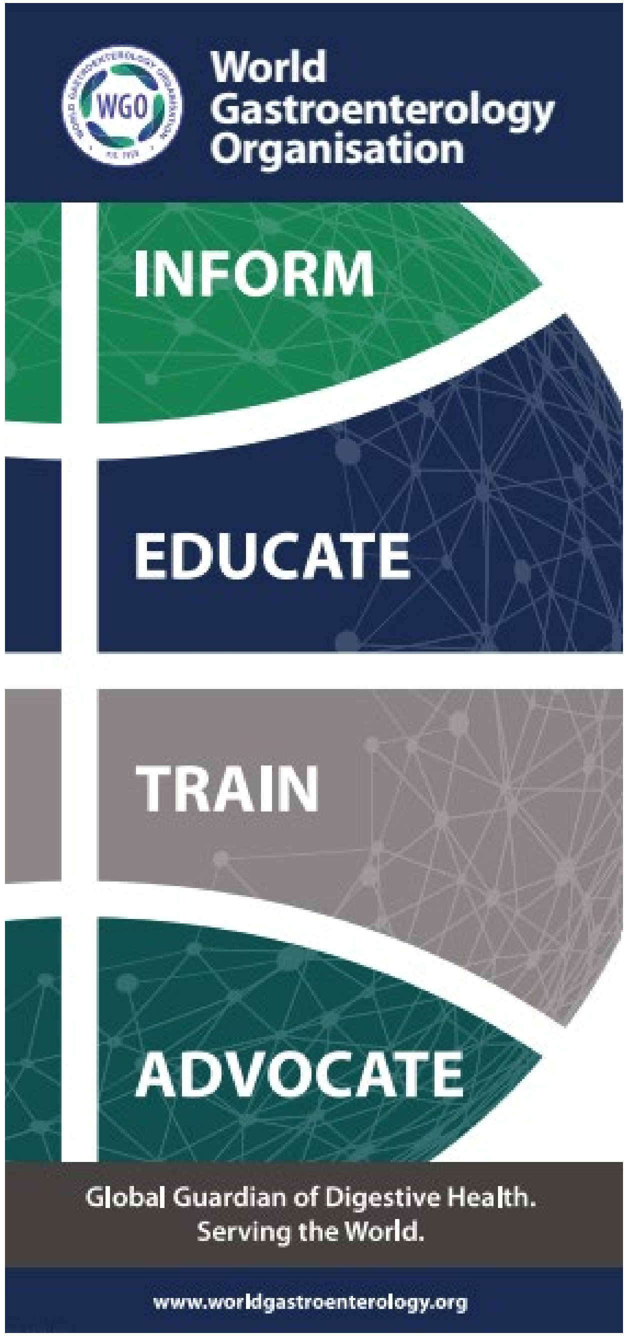
Seeding the Microbiome
Probiotic Prescribing:
Resources to Consider

- Natural Medicines Database
  - Free with IFM Membership
- USProbioticguide.com
  - Free
- Probiotic Advisor
  - Subscription

Seeding the Microbiome Probiotic Prescribing:
Resources to Consider

- World Gastroenterology Organisation Global Guidelines

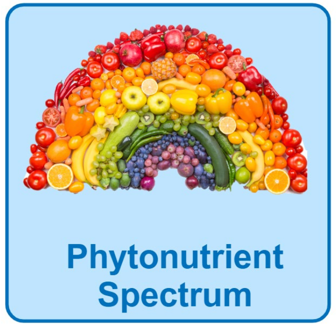
- *https://www.worldgastroenterology.org/guidelines/probiotics-and-prebiotics*



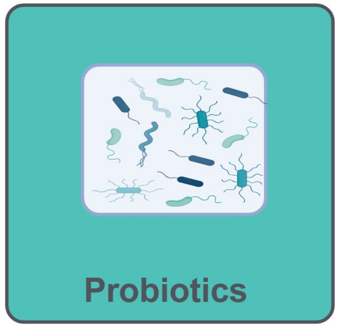
Using the 5R Approach

Anti-Inflammatory Foundation Summary

Dietary



Second-Tier Supplemental Support

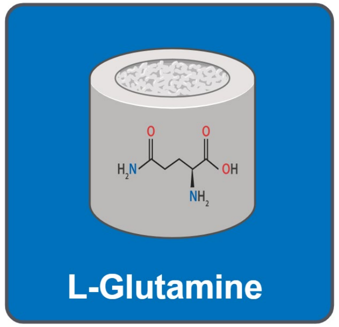


Probiotics: Primarily combinations of two or more Bifidobacterium and Lactobacillus species

Omega-3: 2-4 grams of combined EPA and DHA

Vitamin D: Supplement until optimal blood level of 25(OH)D3 is achieved

- Immune/inflammatory support: **50-80 ng/mL**

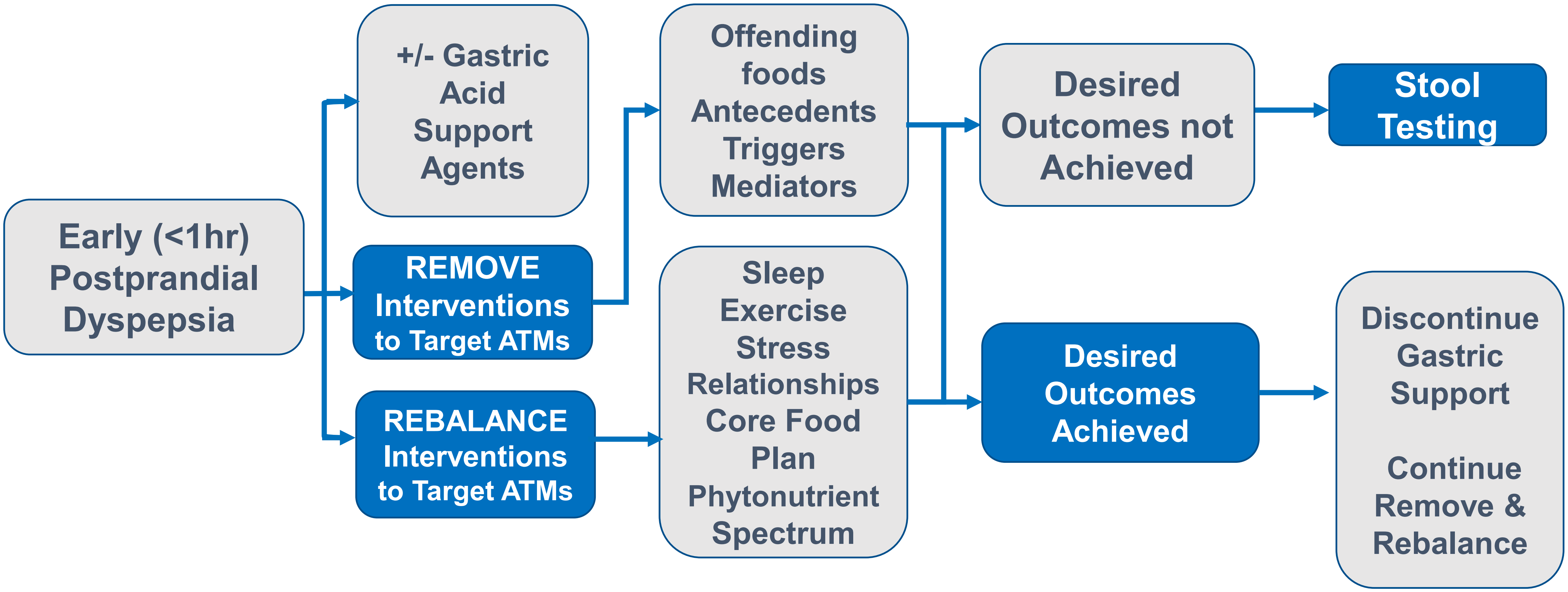


L-glutamine: 5g 3x/day for intestinal support

Long Description: Using the 5 R Approach

Anti-Inflammatory Foundation Summary
Dietary: Core Food Plan and Phytonutrient Spectrum
Second-Tier Supplemental Support: Probiotics, Vitamin D, Omega-3s, L-Glutamine

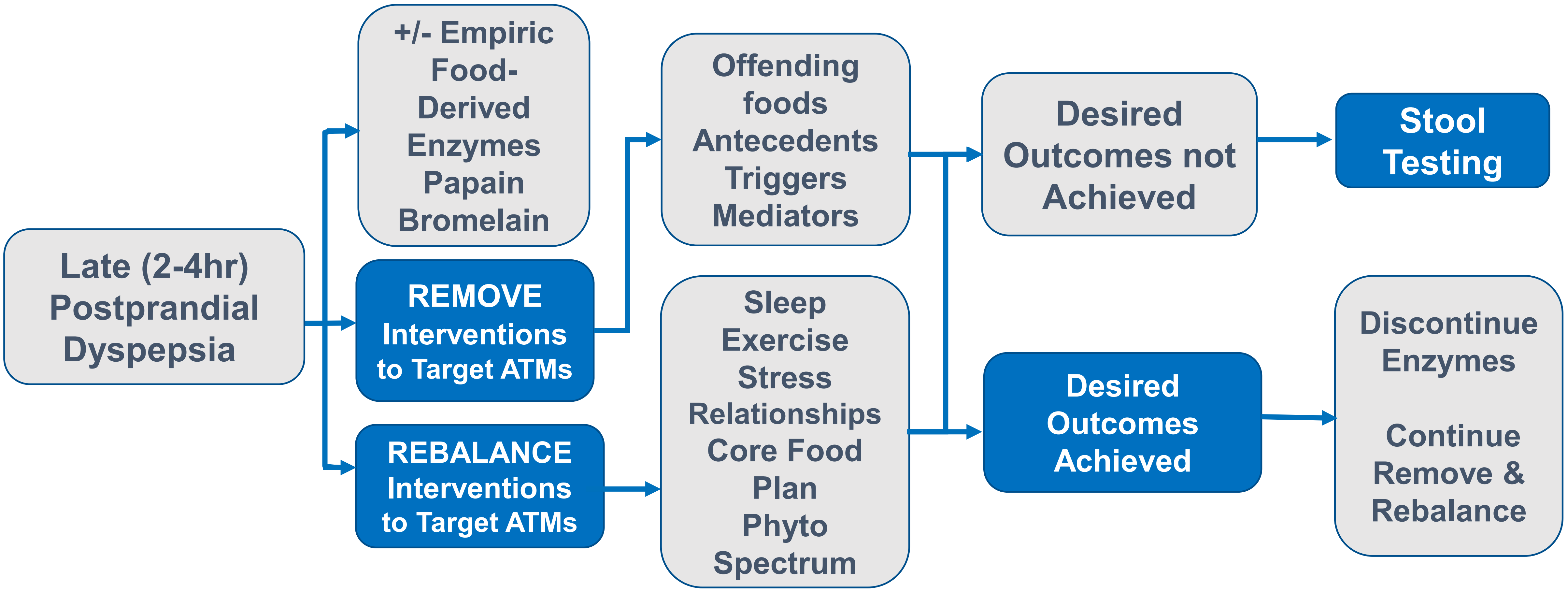
Clinical Flow Path for Early (<1 hour) Postprandial Dyspepsia: Rationale for the Sequence of 5R Approach



Long Description: Clinical Flow Path for Early Postprandial Dyspepsia: 5 R Approach

- Clinical Flow Path for Early (less than 1 hour) Postprandial Dyspepsia
- Early (less than 1 hour) Postprandial Dyspepsia
- Consider gastric acid support agents
- Remove Interventions to target A T M's including offending foods, antecedents, triggers, and mediators
- Rebalance interventions to target A T M's including sleep, exercise, stress, relationships, core food plan, phytonutrient spectrum
- If desired outcomes are achieved, discontinue gastric support and continue to remove and rebalance
- If desired outcomes are not achieved, perform stool testing

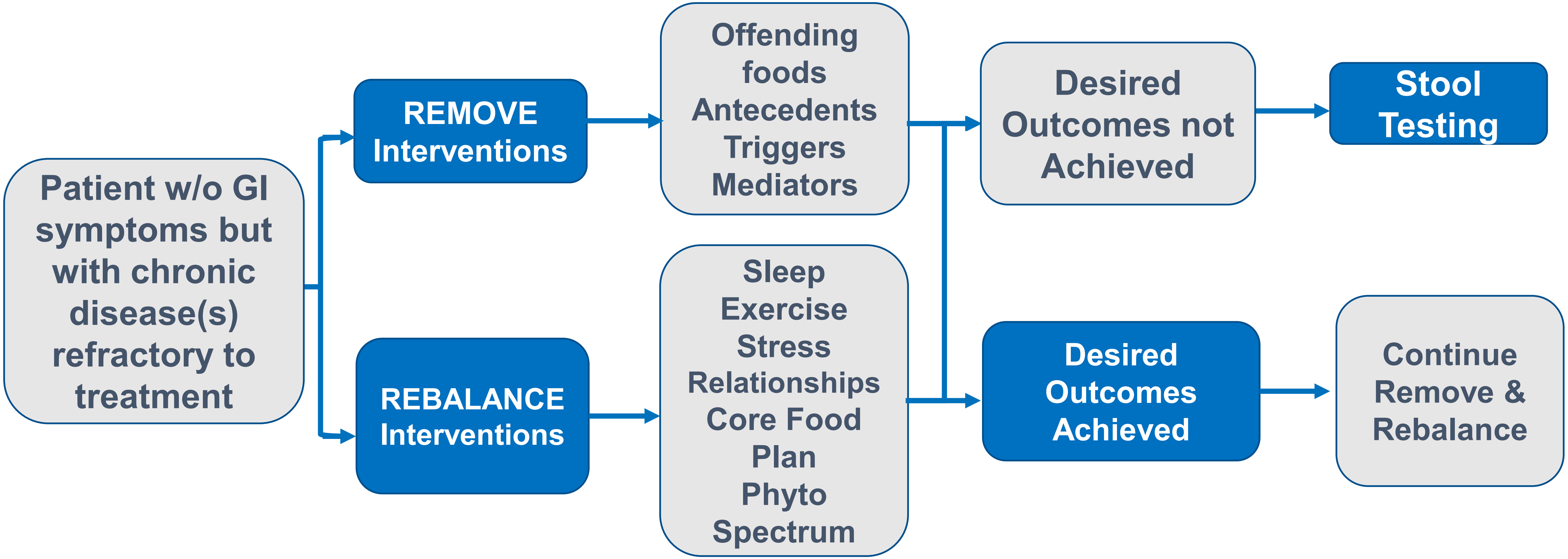
Clinical Flow Path for Late (2-4 hour) Postprandial Dyspepsia: Rationale for the Sequence of 5R Approach



Long Description: Clinical Flow Path for Late Postprandial Dyspepsia: 5 R Approach

- Clinical Flow Path for Late (2-4 hours) Postprandial Dyspepsia
- Late (2-4 hours) Postprandial Dyspepsia
- Consider empiric food-derived enzymes papain and bromelain
- Remove Interventions to target A T M's including offending foods, antecedents, triggers, and mediators
- Rebalance interventions to target A T M's including sleep, exercise, stress, relationships, core food plan, phytonutrient spectrum
- If desired outcomes are achieved, discontinue gastric support and continue to remove and rebalance
- If desired outcomes are not achieved, perform stool testing

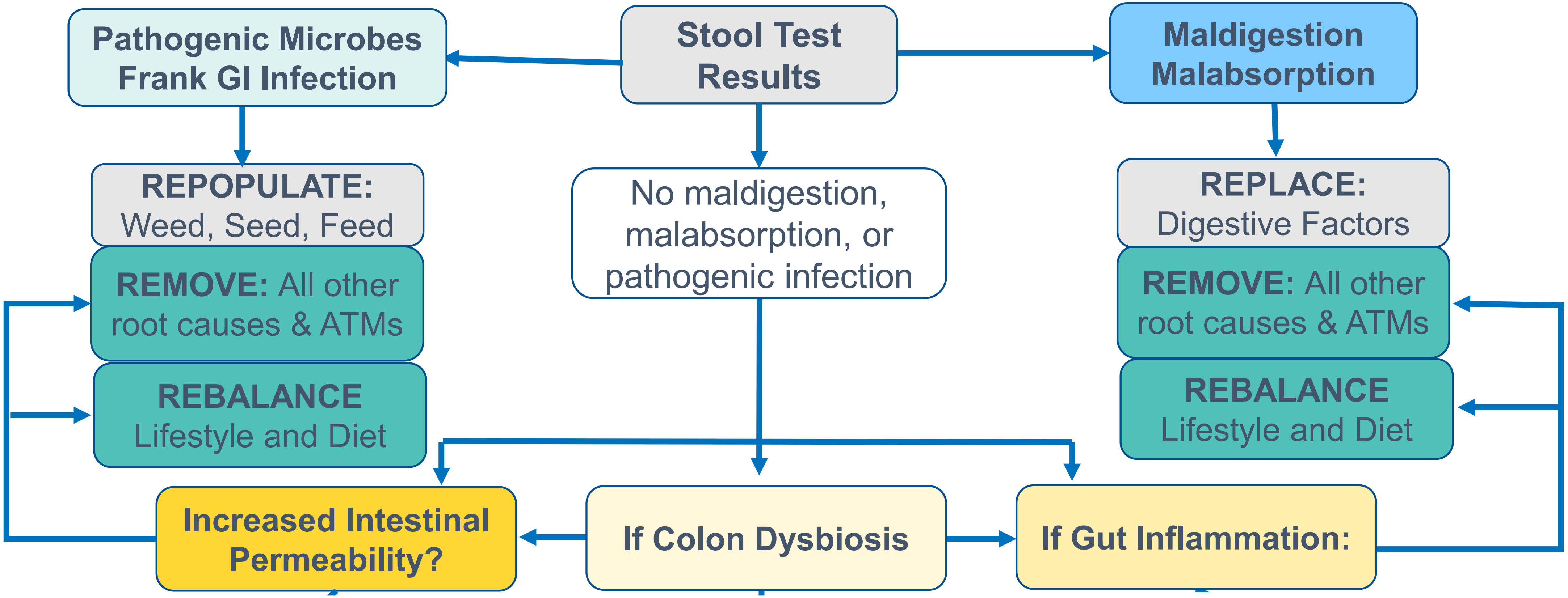
Clinical Flow Path for Patients Without GI Symptoms: Rationale for the Sequence of 5R Approach



Long Description: Clinical Flow Path for Patients Without GI Symptoms: 5 R Approach

- Clinical Flow Path for patients without GI symptoms
- Remove Interventions- offending foods, antecedents, triggers, and mediators
- Rebalance interventions - sleep, exercise, stress, relationships, core food plan, phytonutrient spectrum
- If desired outcomes are achieved, discontinue gastric support and continue to remove and rebalance
- If desired outcomes are not achieved, perform stool testing

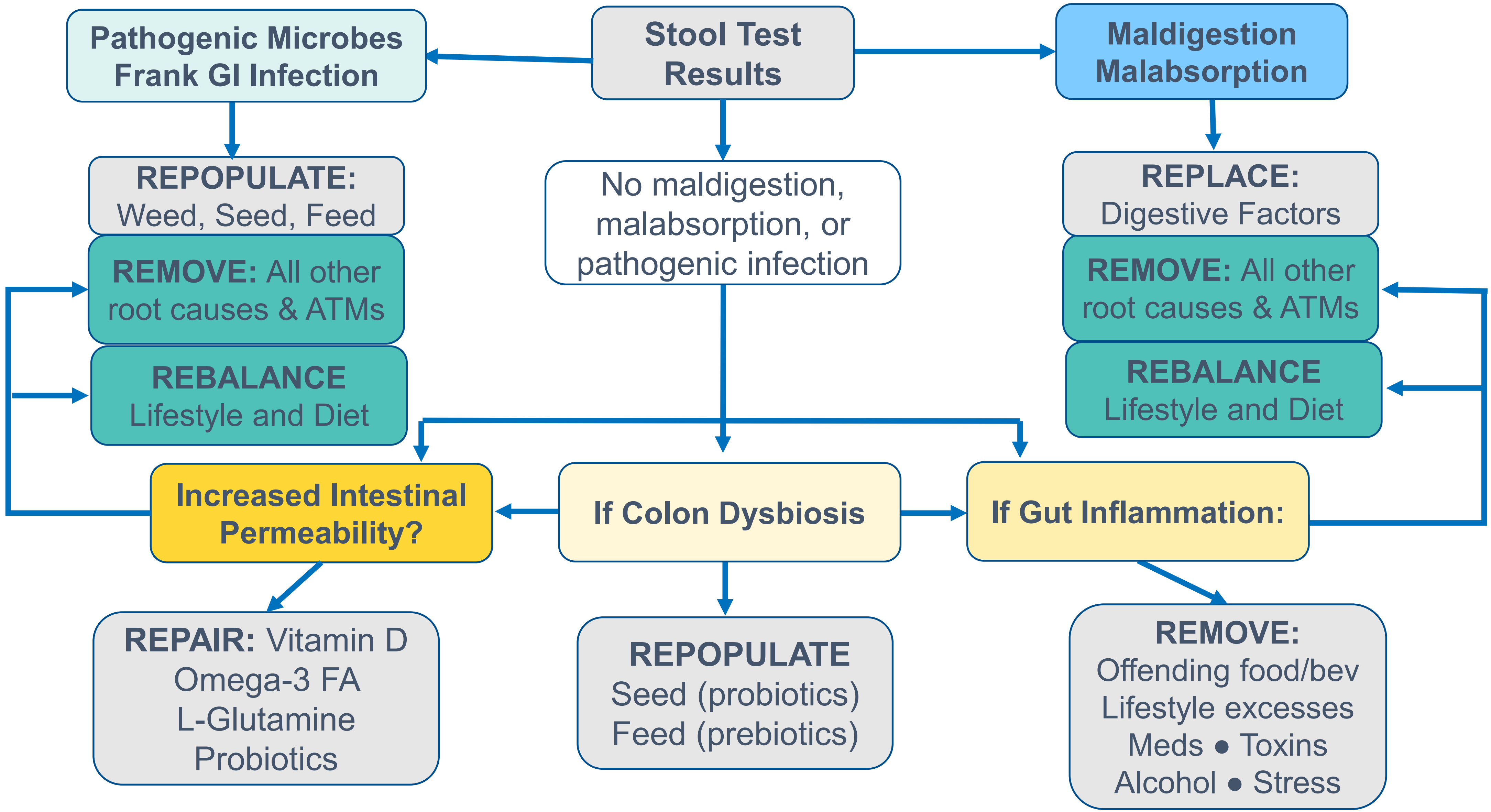
Clinical Flow Path for Stool Test Results: Rationale for the Sequence of 5R Approach



Long Description: Clinical Flow Path for Stool Test Results: 5 R Approach

- If pathogenic microbes or frank GI infection present, then repopulate: weed, seed, feed
- Remove all other root causes and A T Ms
- Rebalance lifestyle and diet
- If maldigestion and malabsorption are present, then replace digestive factors
- Remove all other root causes and A T Ms
- Rebalance lifestyle and diet
- If no maldigestion, malabsorption, or pathogenic infection, is there increased intestinal permeability, colon dysbiosis, or gut inflammation?
- If so,
- Remove all other root causes and A T Ms
- Rebalance lifestyle and diet

Clinical Flow Path for Stool Test Results: Rationale for the Sequence of 5R Approach



Long Description: Clinical Flow Path for Stool Test Results: 5 R Approach

- Adding to the previous flow path, if there is increased intestinal permeability, repair with Vitamin D, omega-3 fatty acids, L-glutamine, and probiotics.
- If there is colon dysbiosis, repopulate with seed probiotics and feed prebiotics.
- If gut inflammation is present, remove offending foods, beverages, lifestyle excesses, medications, toxins, alcohol, and stress.

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Assimilation:

Maldigestion, Malabsorption, Intestinal Permeability, Gut Dysbiosis, Gut Inflammation, Enteric Nervous System, Stool Testing, and 5R Approach

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE